dynamic-networks · investigation report

Dynamic Networks — a catalog of network-reconfiguring rules

running · viva-dynamic-networks · 15 studies · 2026-10-22

Executive summary

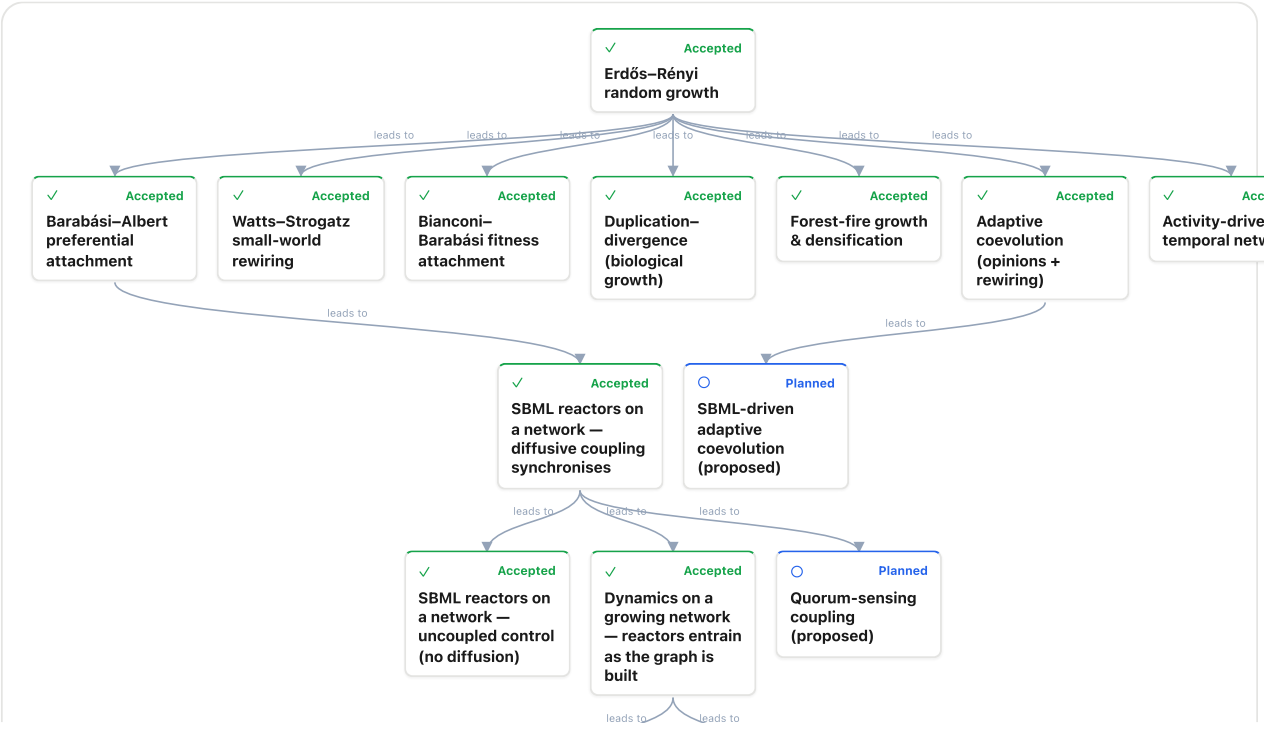
Question. How do different local rules reshape a network over time, and can a single process-bigraph process — driven by pluggable reactions — reproduce the canonical families of network dynamics, each with a clear visualization of the network evolving?

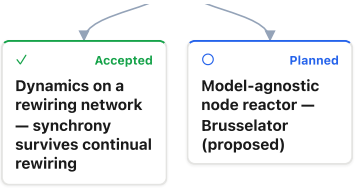
One generalized process reproduces all 8 Family-A dynamics; Family B runs a real SBML Repressilator (via viva-copasi) at every node with inter-node diffusion; and Family C runs that reaction–diffusion ON a topology the process is itself reconfiguring — growing or rewiring — so topology and node state co-evolve.

Across **15** studies — **12** supported · **0** qualified · **0** refuted · **3** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.





Status: ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

▾ 01 Erdős–Rényi random growth

● ACCEPTED

What topology emerges when edges form purely at random, with no preference?

Hypothesis & discriminating control

Random attachment yields a homogeneous Poisson degree distribution with no hubs — the null baseline for every structured rule that follows.

MODEL EXPERIMENT Run the erdos_renyi reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

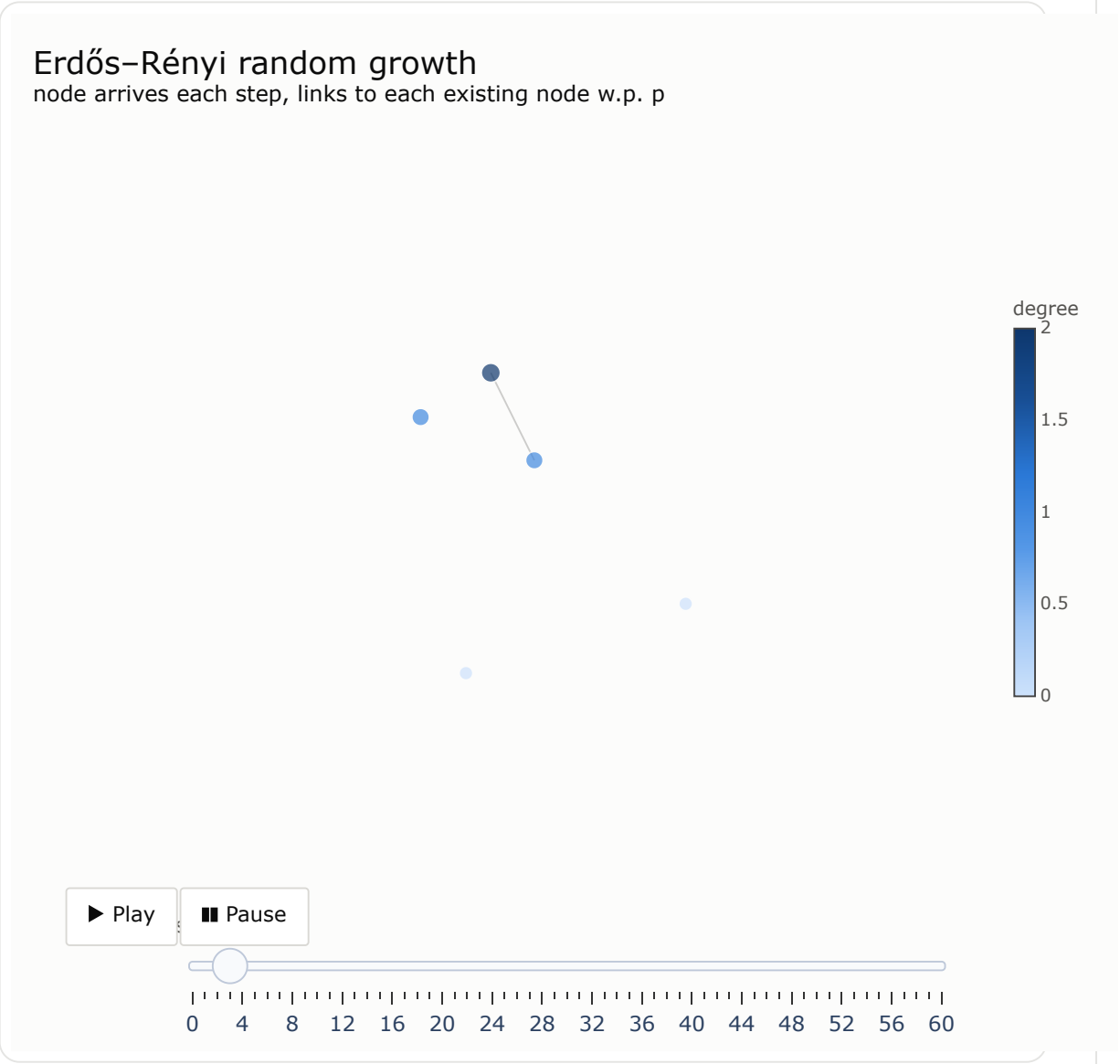
Model & key assumptions

Methods · Simulations

Model `erdos_renyi`

Visualizations

ERDOS RENYI



Finding

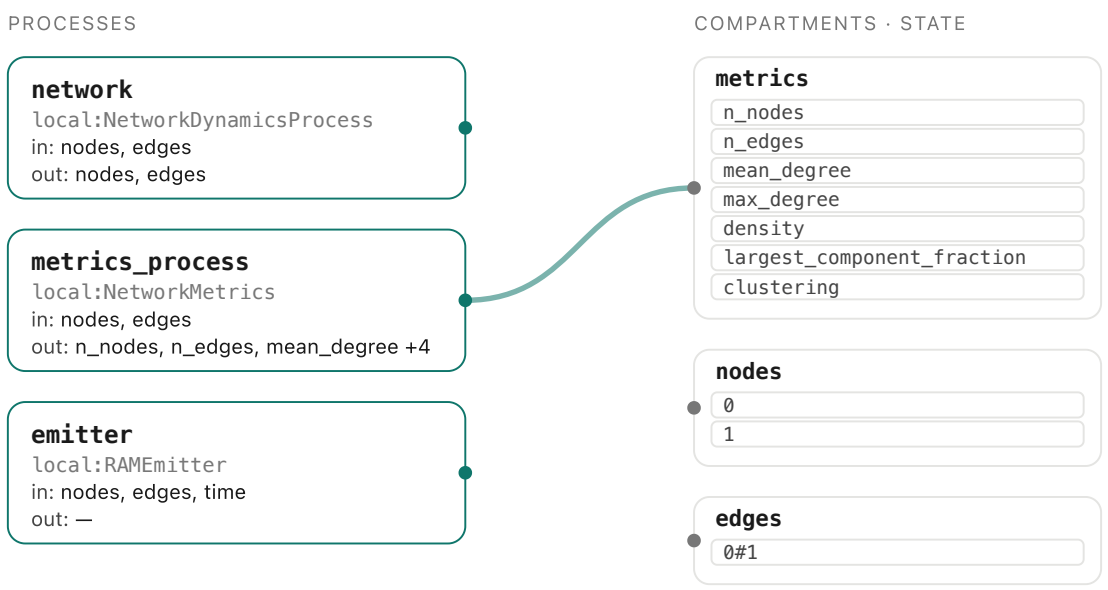
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.erdos_renyi` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.erdos_renyi`

Simulation run

Reproduce: `vwb run study 01-erdos-renyi-random-growth`

Tests

A giant component spans ≥90% of nodes at the end. `er-giant-component · quantitative · largest_component_fraction`	`at_least 0.9`	pending
Max degree is modest (< 4× mean degree) — no scale-free hub. `er-no-dominant-hub · quantitative · max_degree`	`at_most 16`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 02 Barabási–Albert preferential attachment ● ACCEPTED

Does degree-proportional attachment produce scale-free hubs?

Hypothesis & discriminating control

Linking new nodes preferentially to high-degree nodes yields a power-law degree distribution with a few dominant hubs.

MODEL EXPERIMENT Run the preferential_attachment reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

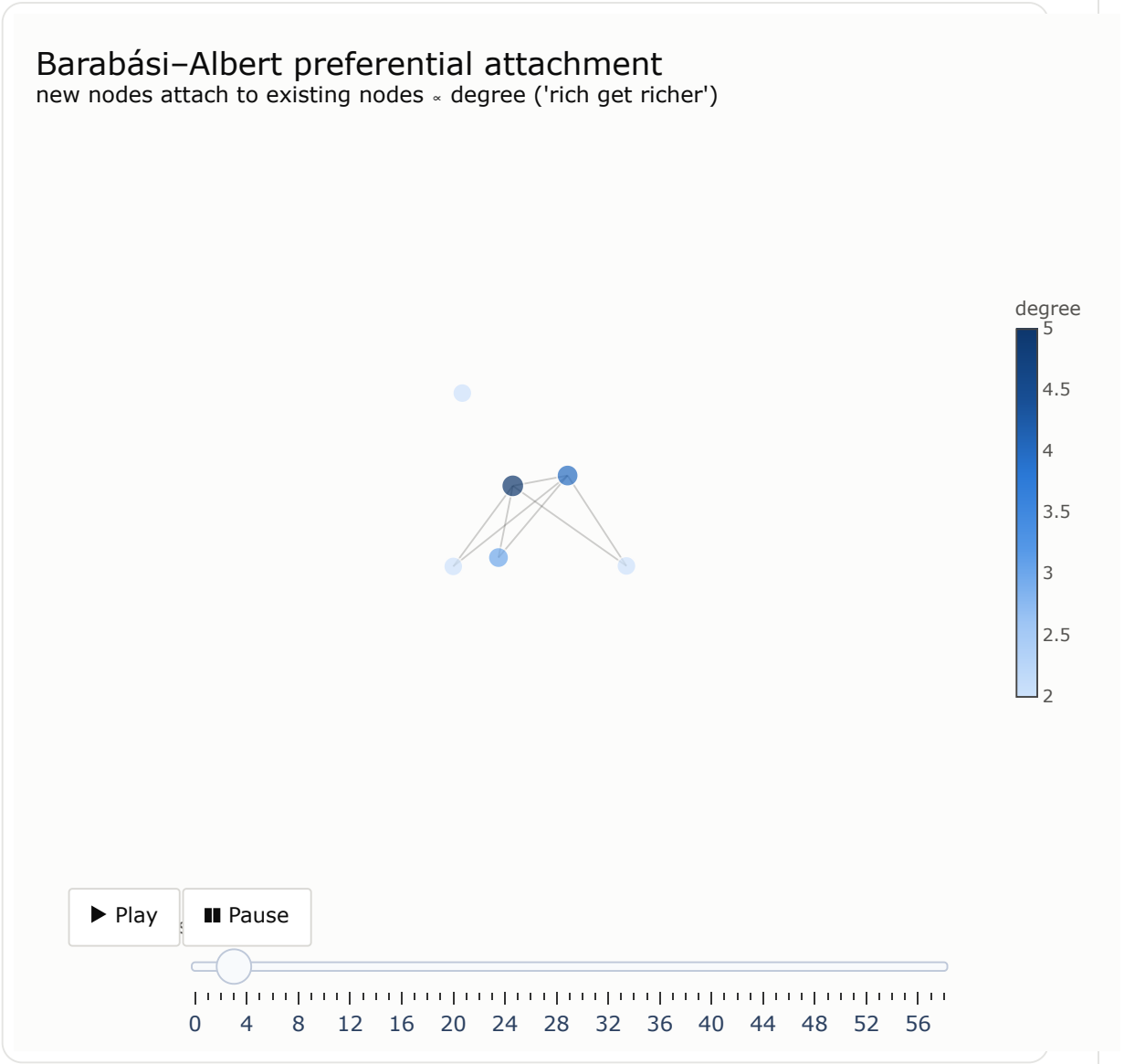
Model & key assumptions

Methods · Simulations

Model `preferential_attachment`

Visualizations

PREFERENTIAL ATTACHMENT



Finding

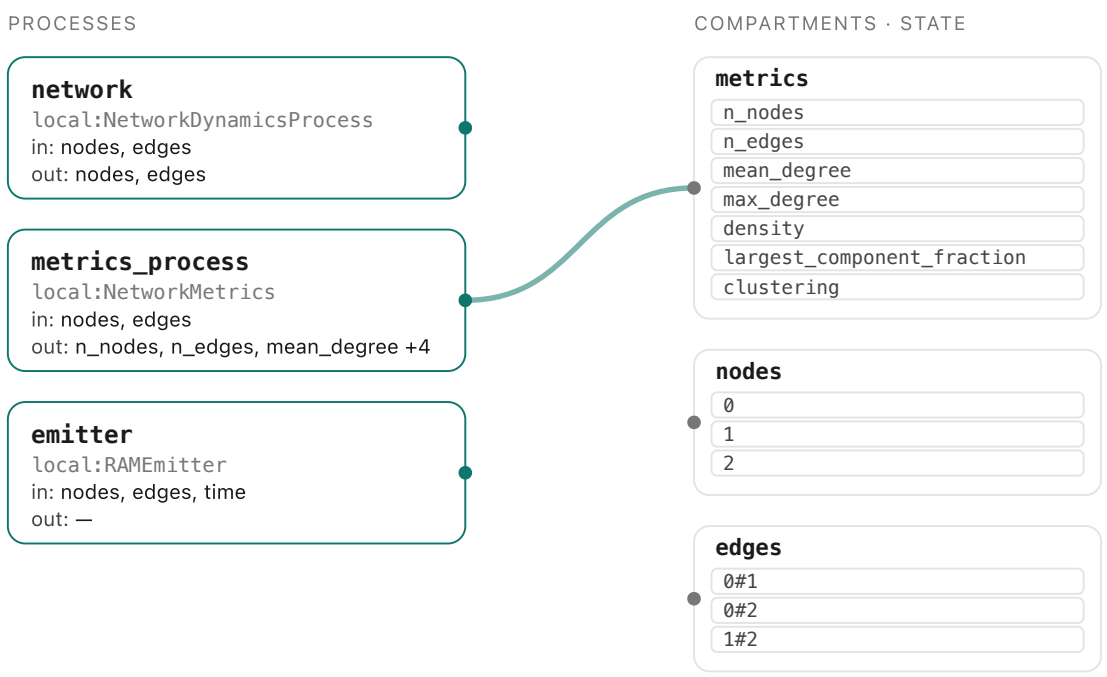
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.preferential_attachment` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.preferential_attachment`

Simulation run

Reproduce: `vwb run study 02-preferential-attachment`

Tests

The maximum degree reaches ≥10 (a dominant hub). `ba-hub-emerges · quantitative · max_degree`	`at_least 10`	pending
The network stays fully connected (BA growth never disconnects). `ba-connected · quantitative · largest_component_fraction`	`at_least 0.99`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 03 Watts–Strogatz small-world rewiring

● ACCEPTED

How does random rewiring of a ring lattice change its structure?

Hypothesis & discriminating control

Rewiring a small fraction of lattice edges collapses path length while clustering stays high — the small-world regime — without changing node or edge counts.

MODEL EXPERIMENT Run the small_world_rewire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

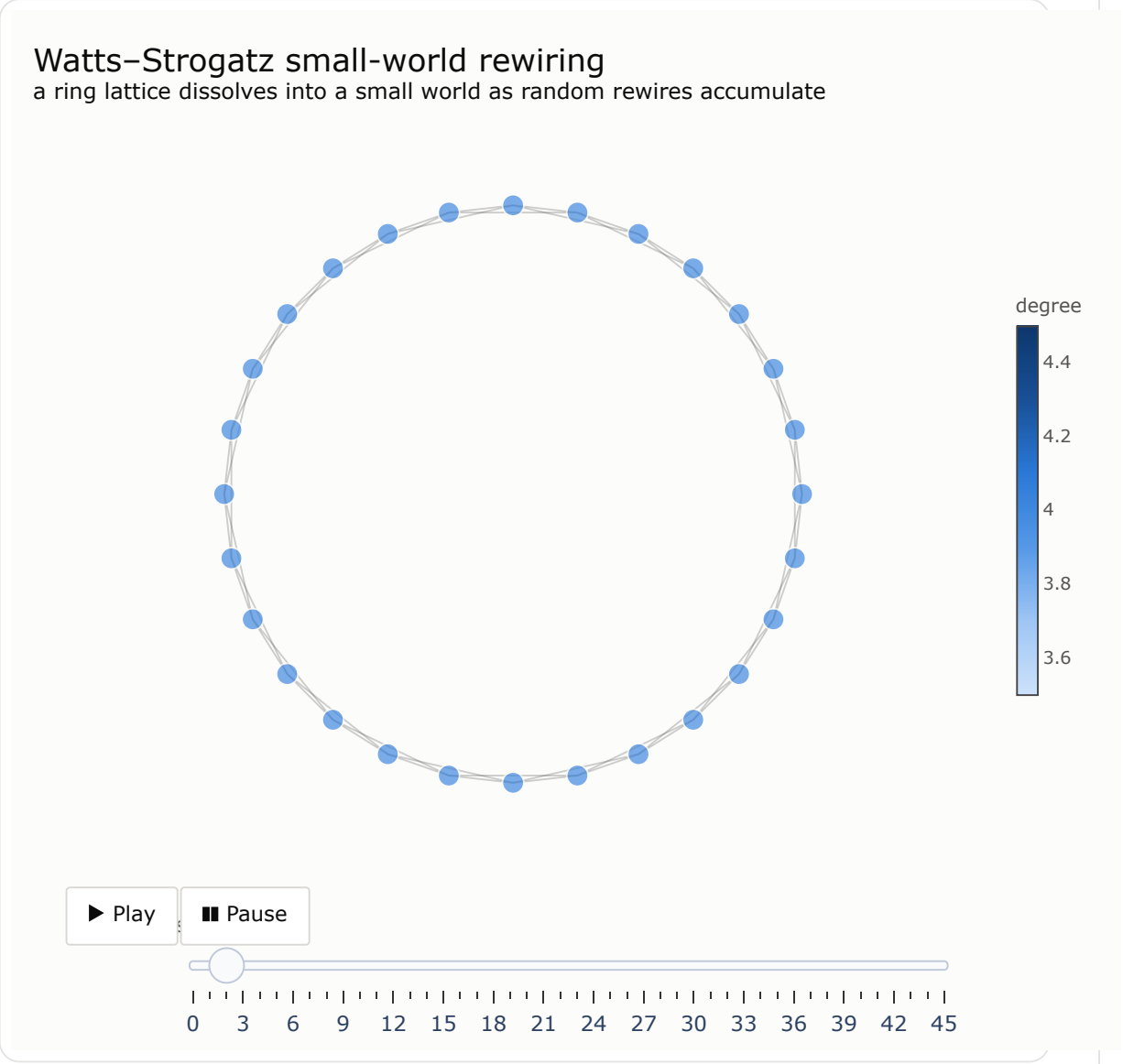
Model & key assumptions

Methods · Simulations

Model `small_world_rewire`

Visualizations

SMALL WORLD REWIRE



Finding

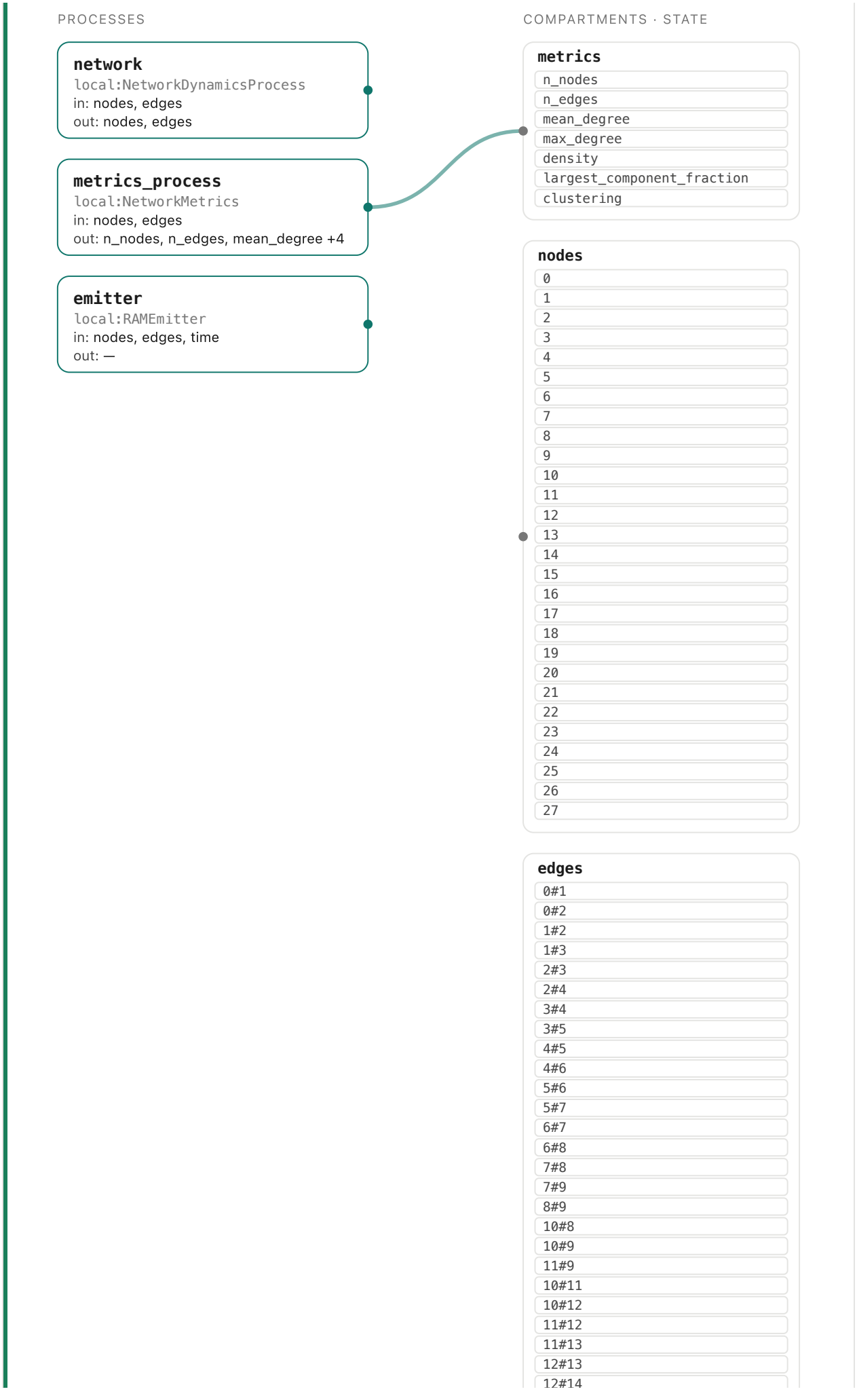
finding · Plausible

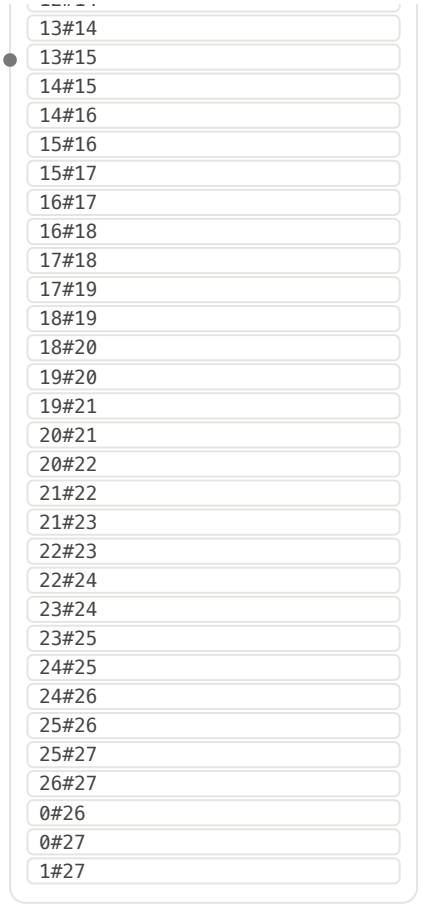
Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals


PROCESSES
network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges
metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4
emitter
local:RAMEmitter
in: nodes, edges, time
out: —
COMPARTMENTS · STATE
metrics
n_nodes
n_edges
mean_degree
max_degree
density
largest_component_fraction
clustering
nodes
0
1
2
3
4
5
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Static bigraph of `viva_dynamic_networks.composites.family_a.small_world_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.small_world_rewire`

Simulation run

Reproduce: `vwb run study 03-small-world-rewiring`

Tests

Edge count equals the initial ring-lattice edge count at the end. `sw-edge-count-conserved · invariant · n_edges`	`equals 56`	pending
Node count is unchanged (no growth). `sw-node-count-conserved · invariant · n_nodes`	`equals 28`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 04 Bianconi–Barabási fitness attachment

● ACCEPTED

Can an intrinsic node fitness let a latecomer overtake older, higher-degree nodes?

Hypothesis & discriminating control

Weighting attachment by fitness×degree still produces hubs, but the leading hub need not be the oldest node — fitness breaks the pure first-mover advantage.

MODEL EXPERIMENT Run the fitness_attachment reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

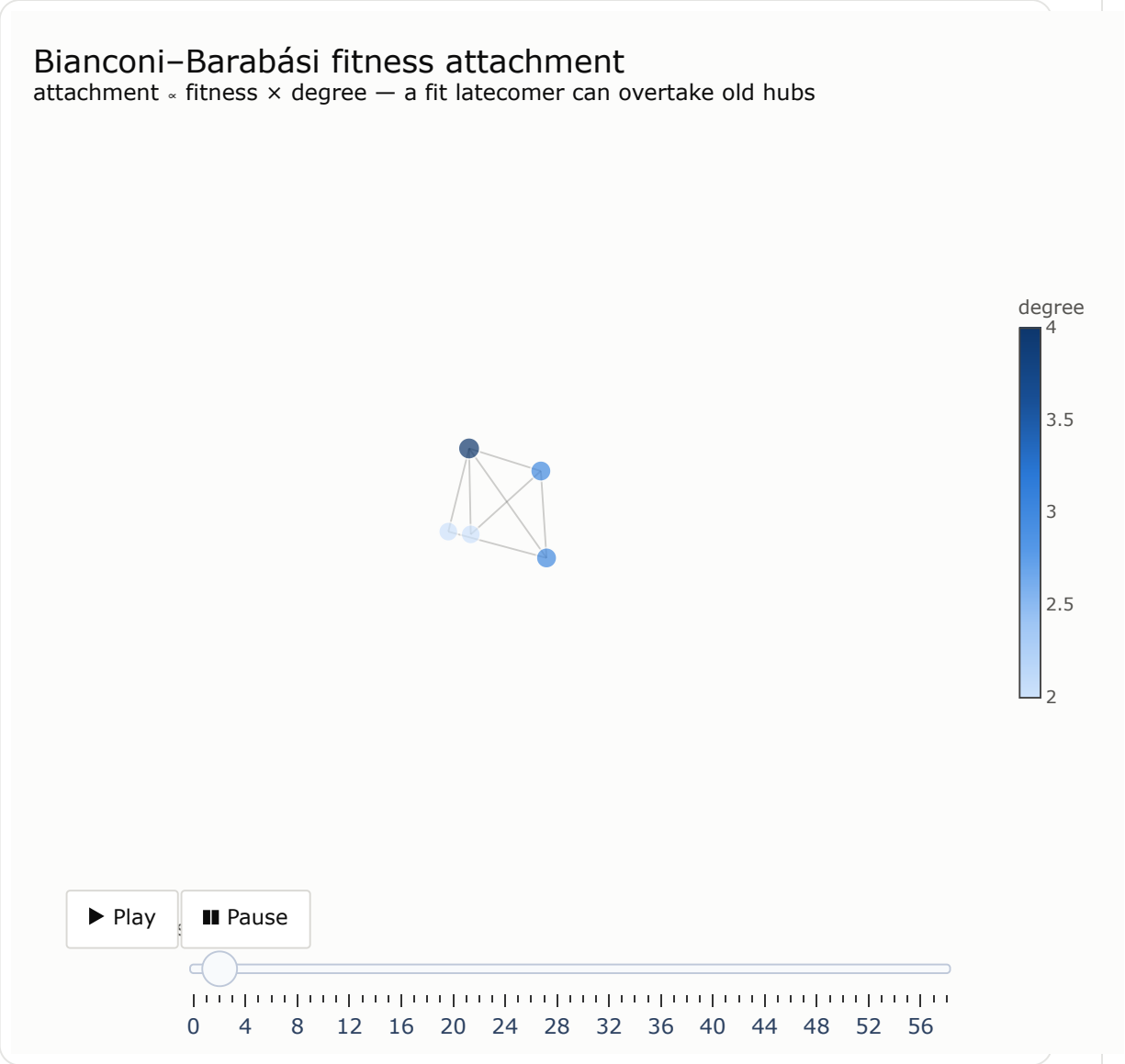
Model & key assumptions

Methods · Simulations

Model `fitness_attachment`

Visualizations

FITNESS ATTACHMENT



Finding

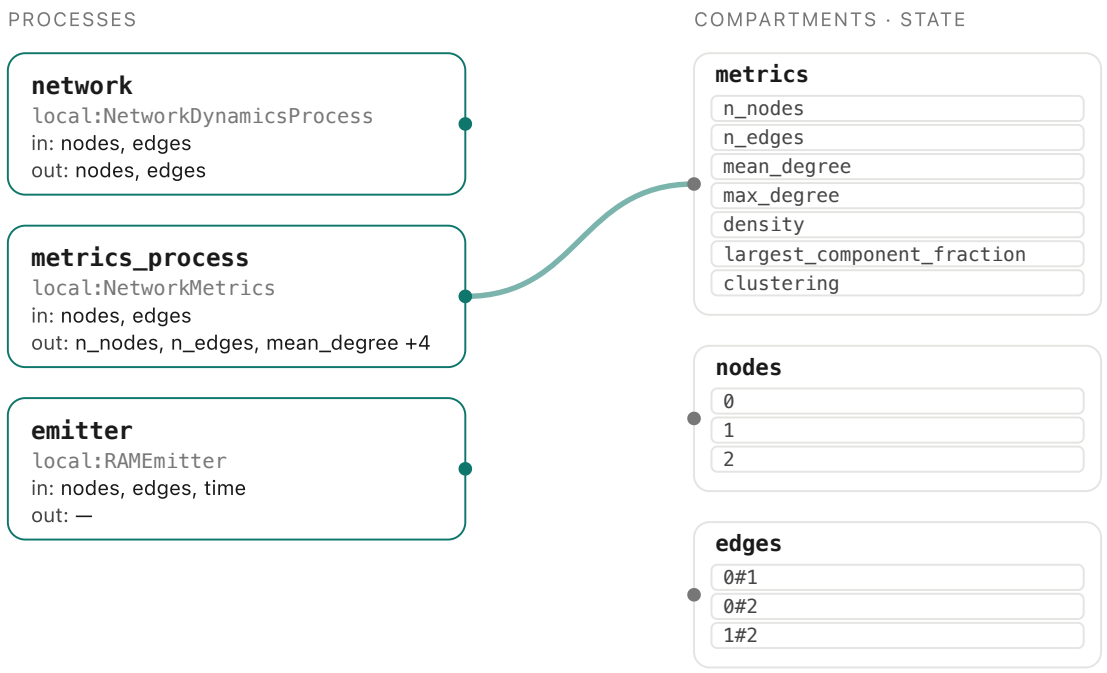
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.fitness_attachment` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.fitness_attachment`

Simulation run

Reproduce: `vwb run study 04-fitness-condensation`

Tests

A dominant hub forms (max degree ≥10). `fit-hub-emerges · quantitative · max_degree`	`at_least 10`	pending
The network stays connected. `fit-connected · quantitative · largest_component_fraction`	`at_least 0.99`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 05 Duplication–divergence (biological growth)

● ACCEPTED

Does genome-style duplication-and-divergence reproduce protein-network topology?

Hypothesis & discriminating control

Copying a random node's neighbourhood and pruning edges yields a heavy-tailed, locally clustered network like observed protein-interaction graphs.

MODEL EXPERIMENT Run the duplication_divergence reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

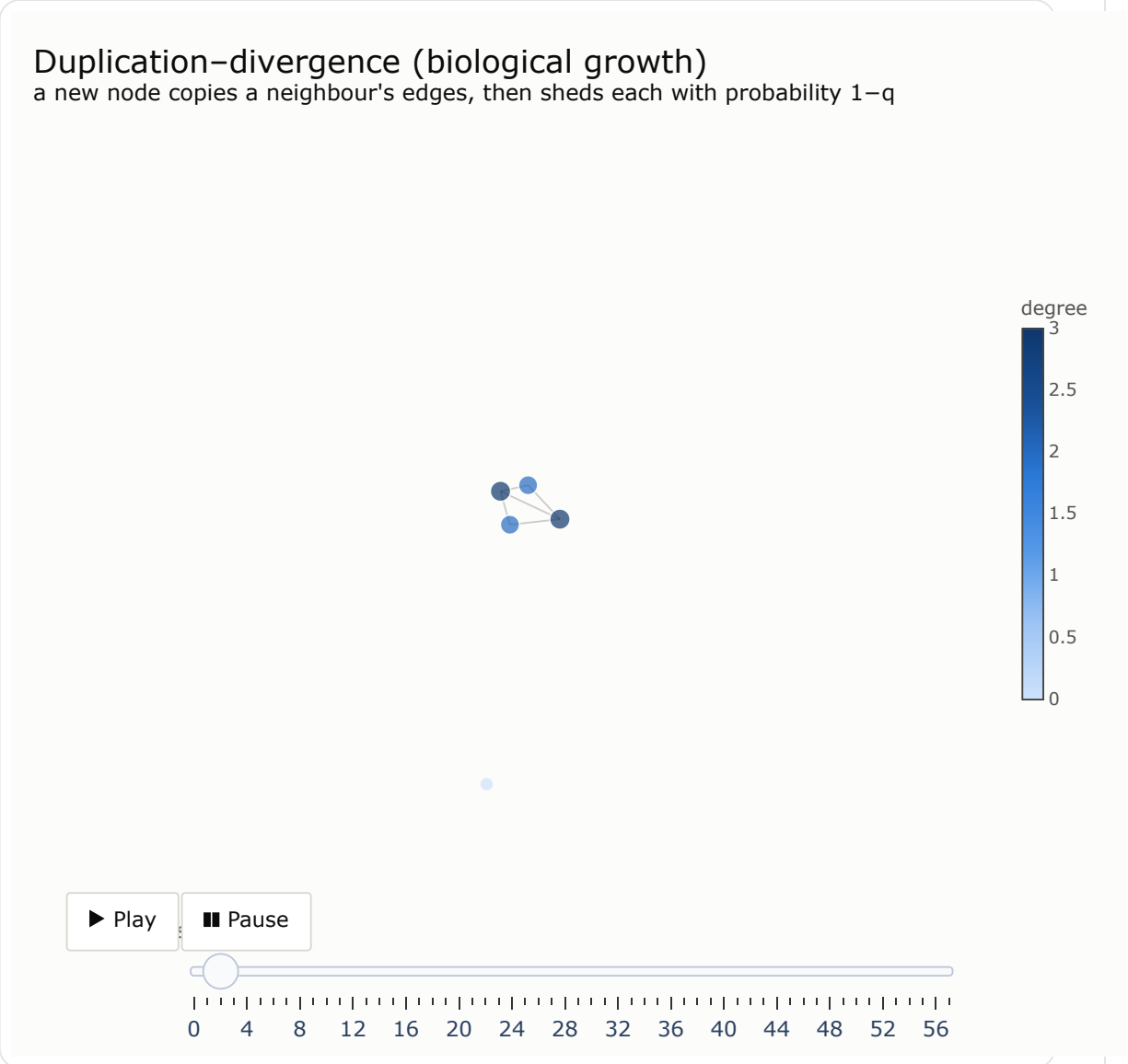
Model & key assumptions

Methods · Simulations

Model `duplication_divergence`

Visualizations

DUPLICATION DIVERGENCE



Finding

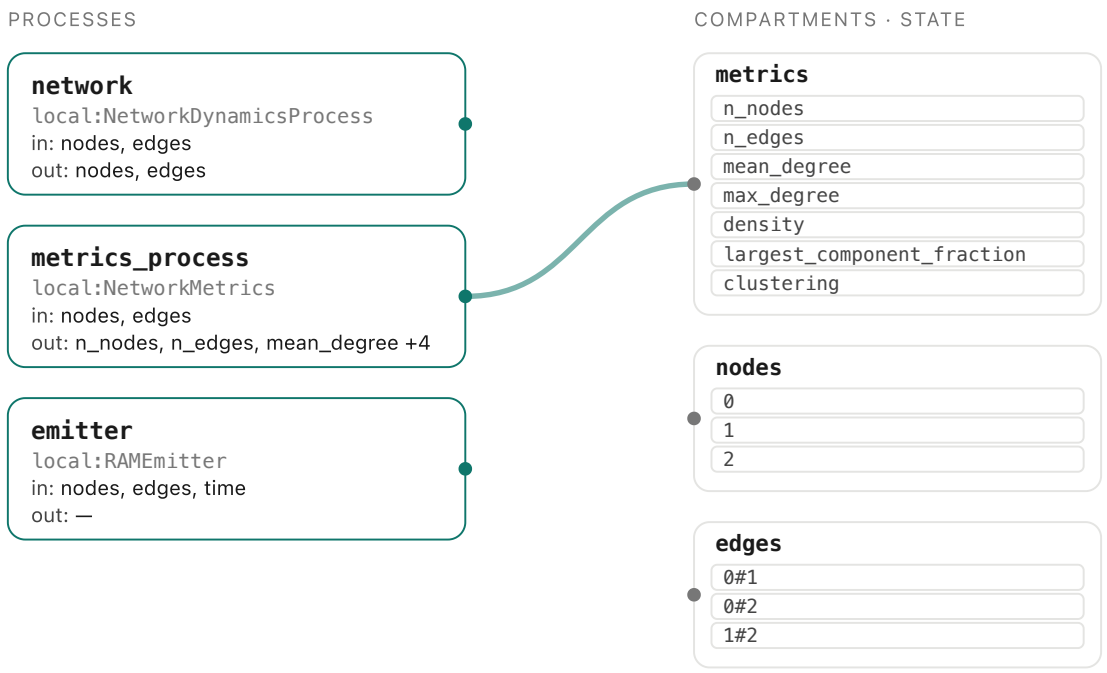
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.duplication_divergence` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.duplication_divergence`

Simulation run

Reproduce: `vwb run study 05-duplication-divergence`

Tests

A heavy-tailed hub forms (max degree ≥6). `dd-hub-forms · quantitative · max_degree`	`at_least 6`	pending
Average clustering is non-trivial (≥0.1). `dd-clustered · quantitative · clustering`	`at_least 0.1`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 06 **Forest-fire growth & densification** ● ACCEPTED

Does recursive forest-fire burning densify the network over time?

Hypothesis & discriminating control

Because each arrival can ignite many links, the edge count grows super-linearly in the node count — the network densifies and its mean degree rises.

MODEL EXPERIMENT Run the forest_fire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

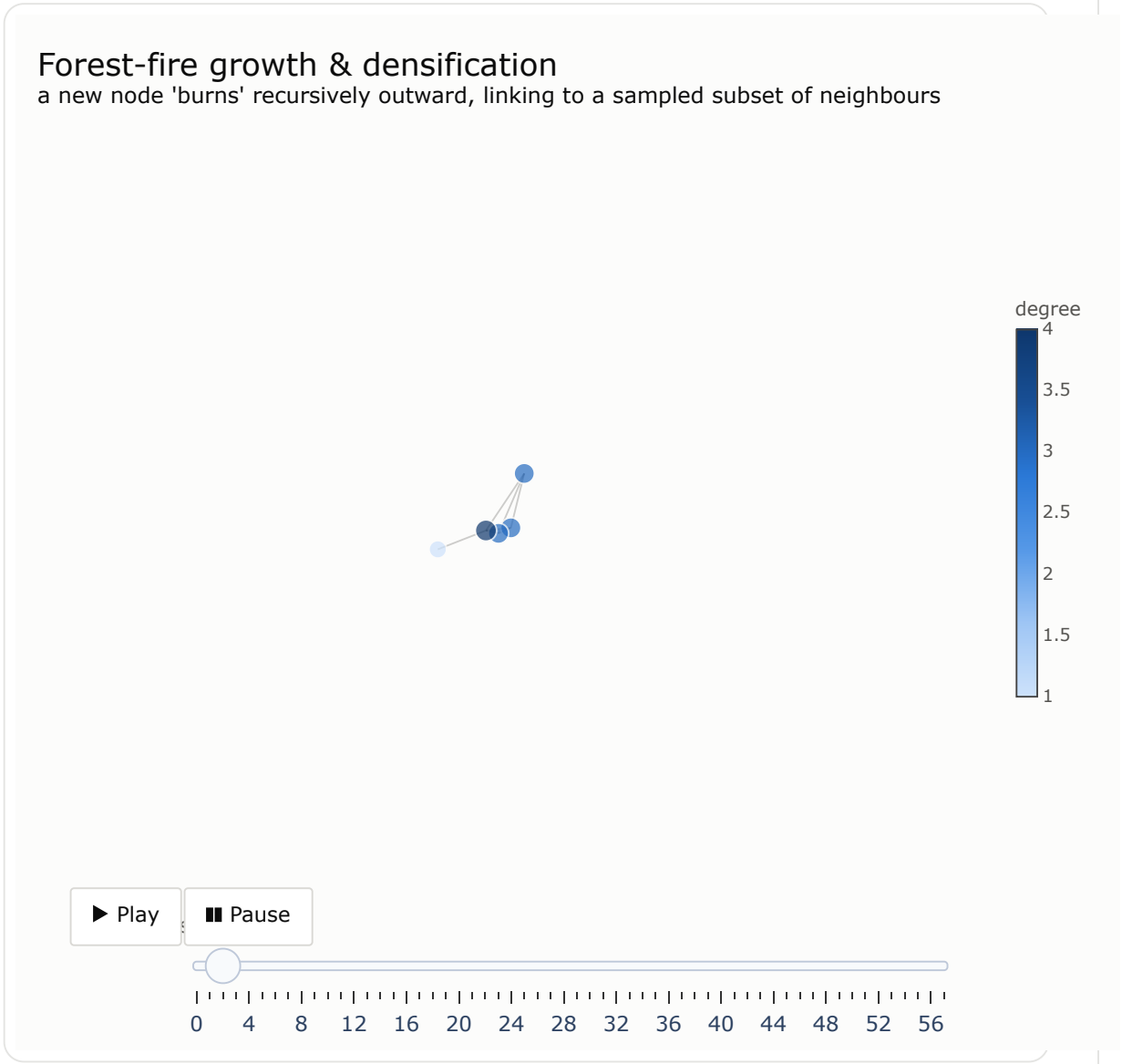
Model & key assumptions

Methods · Simulations

Model `forest_fire`

Visualizations

FOREST FIRE



Finding

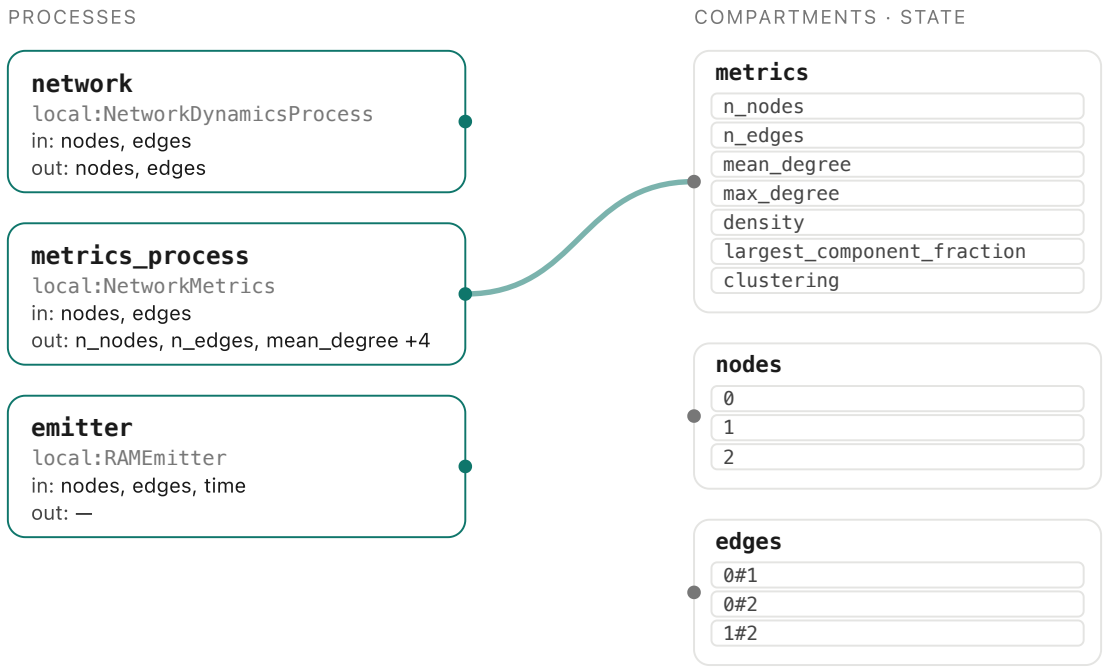
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.forest_fire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.forest_fire`

Simulation run

Reproduce: `vwb run study 06-forest-fire-densification`

Tests

Mean degree increases from start to end (densification). `ff-densification · quantitative · mean_degree`	`at_least 3`	pending
The network stays connected. `ff-connected · quantitative · largest_component_fraction`	`at_least 0.99`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 07 Adaptive coevolution (opinions + rewiring)

● ACCEPTED

When topology and node state co-adapt, does the network organise into like-minded communities?

Hypothesis & discriminating control

Letting nodes rewire away from differing neighbours and toward similar ones (while opinions also spread) raises assortativity and segregates the two opinions — the coevolution bridge between Family A and dynamics-on-networks.

MODEL EXPERIMENT Run the adaptive_rewire reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

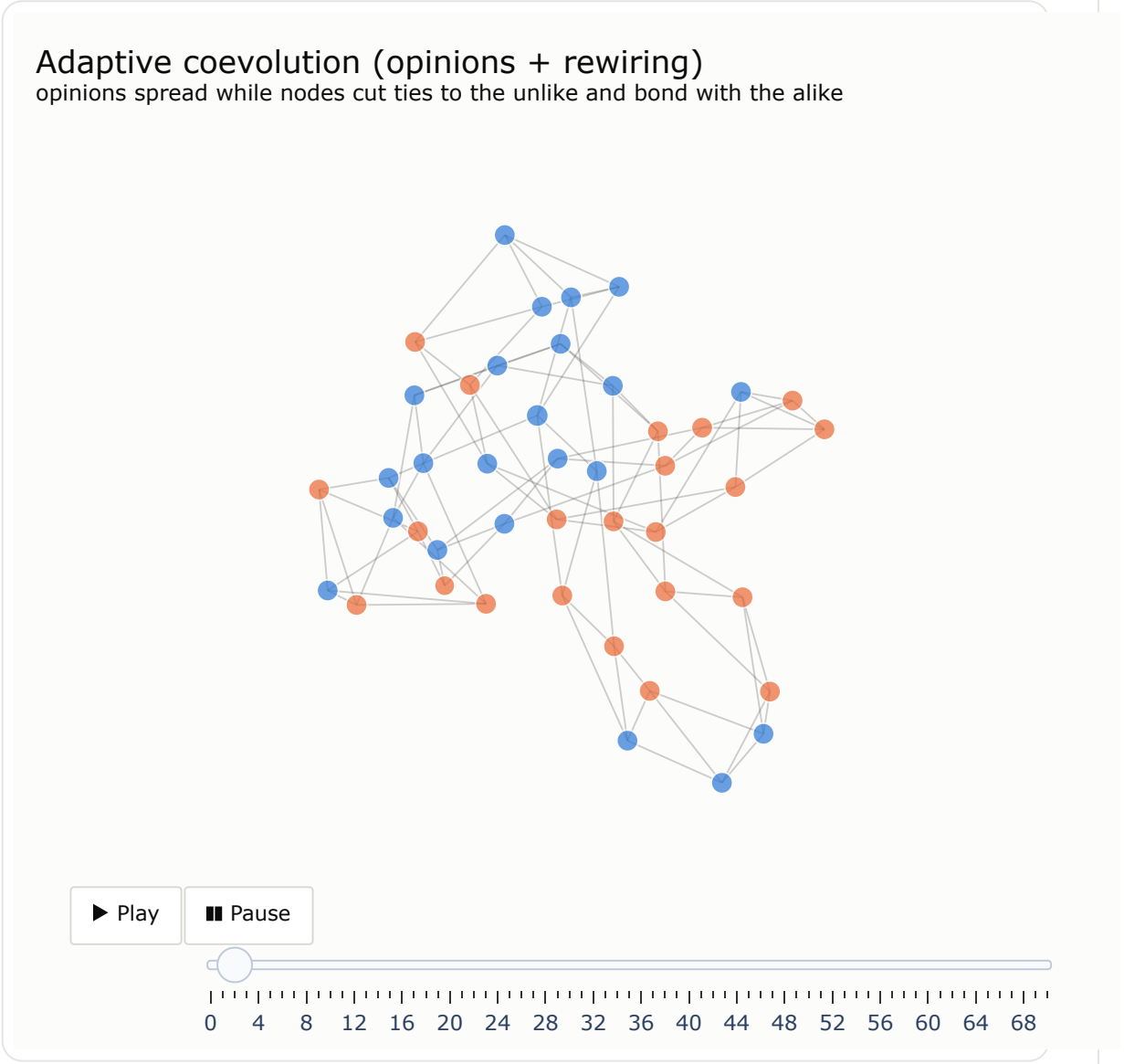
Model & key assumptions

Methods · Simulations

Model `adaptive_rewire`

Visualizations

ADAPTIVE REWIRE



Finding

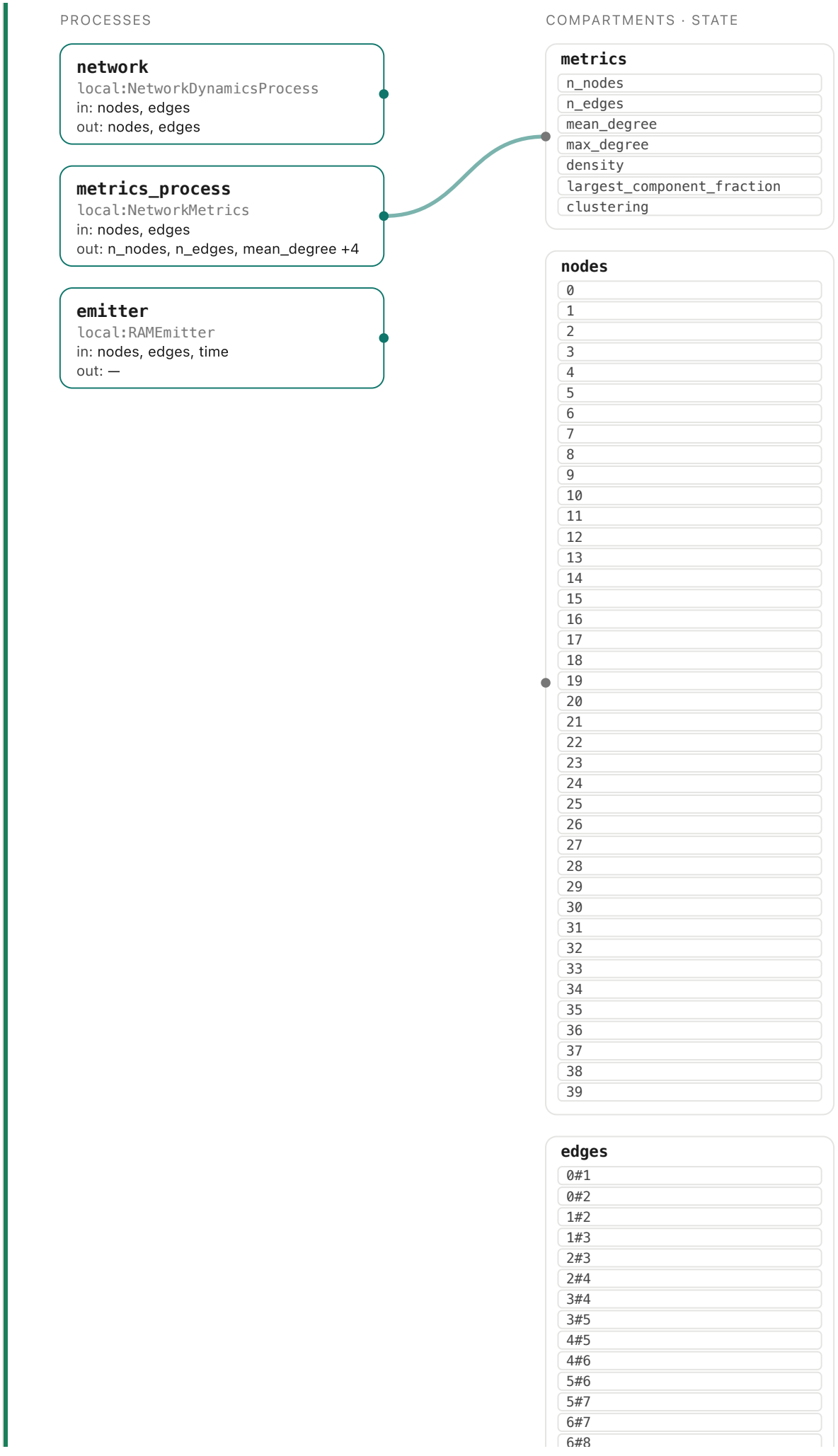
finding Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals


PROCESSES
network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges
metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4
emitter
local:RAMEmitter
in: nodes, edges, time
out: —
COMPARTMENTS · STATE
metrics
n_nodes
n_edges
mean_degree
max_degree
density
largest_component_fraction
clustering
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Static bigraph of `viva_dynamic_networks.composites.family_a.adaptive_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.adaptive_rewire`

Simulation run

Reproduce: `vwb run study 07-adaptive-coevolution`

Tests

Edge count is conserved (rewiring, not growth). `ad-edge-count-conserved · invariant · n_edges`	`equals 88`	pending
Node count is conserved. `ad-node-count-conserved · invariant · n_nodes`	`equals 44`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 08 Activity-driven temporal network

● ACCEPTED

What does an activity-driven contact sequence look like, where edges last only one time window?

Hypothesis & discriminating control

With edges reset every step and nodes firing in proportion to their activity, each instantaneous graph is sparse and blinking, while the time-integrated degree follows the activity distribution.

MODEL EXPERIMENT Run the activity_driven reaction and show how it reconfigures the network, measuring its signature structural observables.

OBSERVABLE `2 acceptance tests`

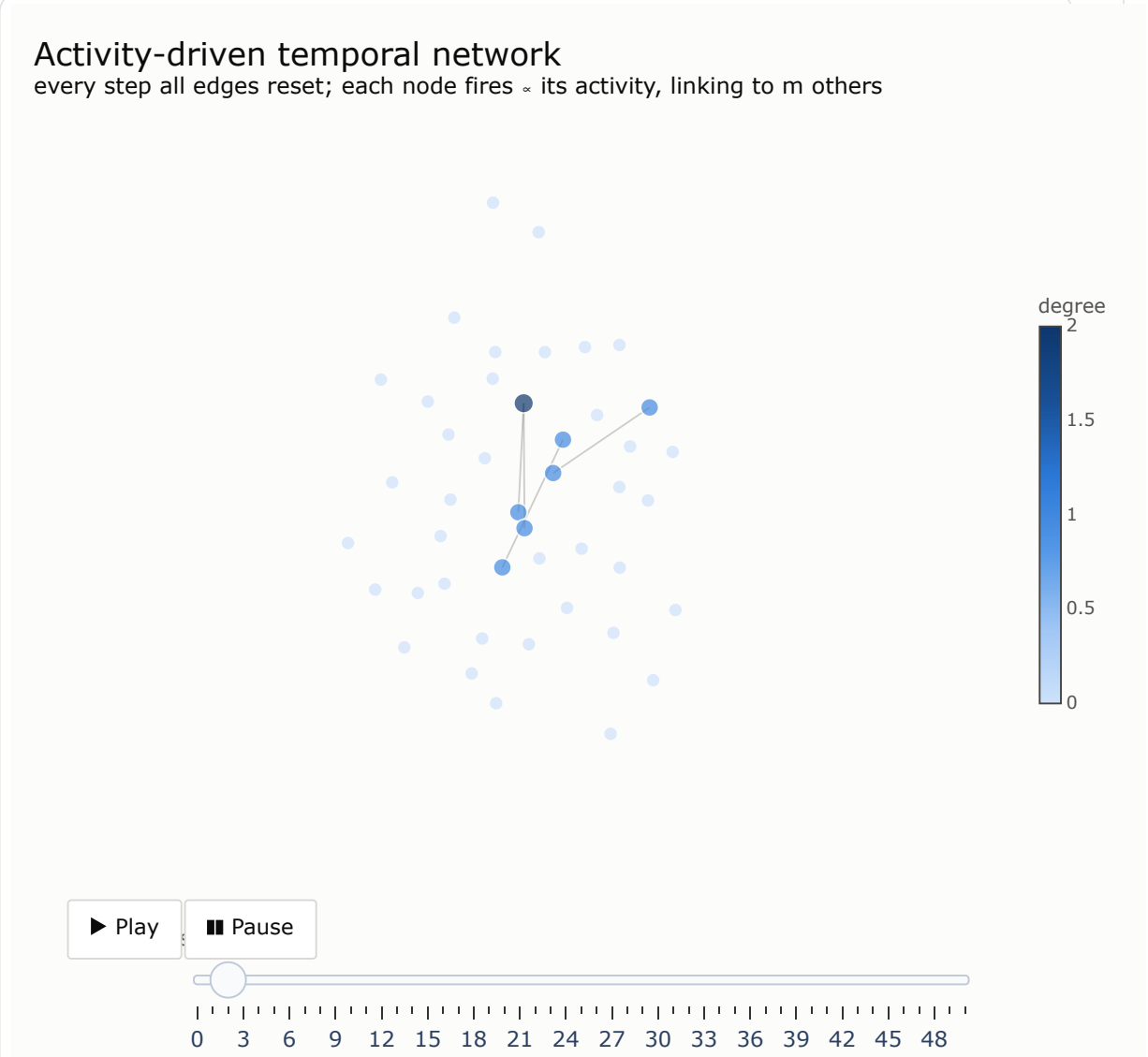
Model & key assumptions

Methods · Simulations

Model `activity_driven`

Visualizations

ACTIVITY DRIVEN



Finding

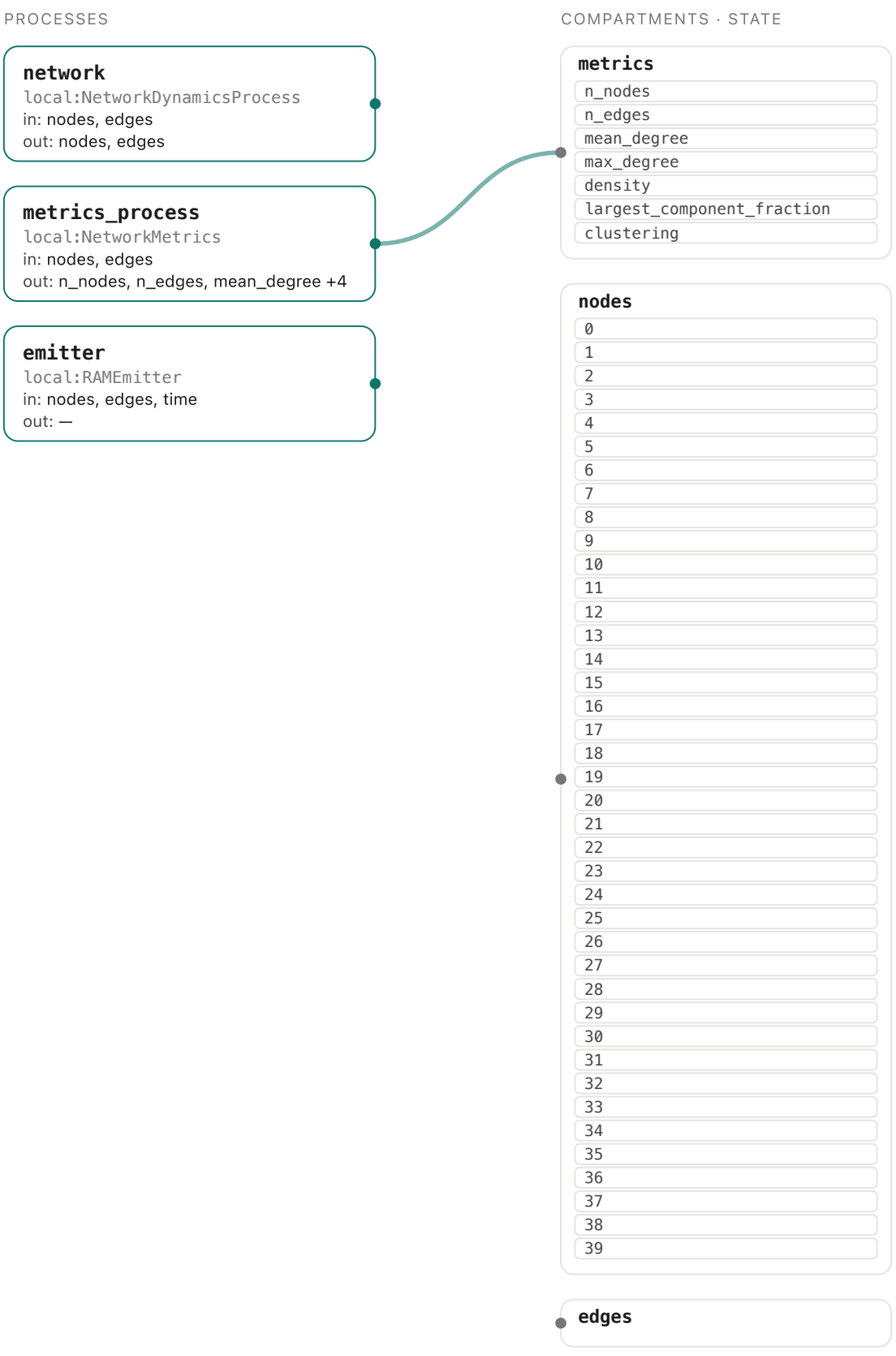
finding · Plausible

Verdict

● PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_a.activity_driven` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.activity_driven`

Simulation run

Reproduce: `vwb run study 08-activity-driven-temporal`

Tests

Node count is constant at 44.
`adn-node-count-conserved · invariant · n_nodes`

`equals 44` pending

Instantaneous snapshots stay sparse (mean degree < 2).
`adn-sparse-snapshot · quantitative · mean_degree`

`at_most 2` pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 09

SBML reactors on a network — diffusive coupling synchronises

● ACCEPTED

When a continuous SBML oscillator runs at every node and species diffuse along the edges, does the network synchronise?

Hypothesis & discriminating control

Diffusive coupling of per-node Repressilators pulls their phases together, so the across-node spread of a protein collapses toward zero — synchronisation.

MODEL EXPERIMENT Run the repressilator_network_coupled composite (SBML reactors + inter-node transport) and measure across-node synchrony.

OBSERVABLE `1 acceptance test`

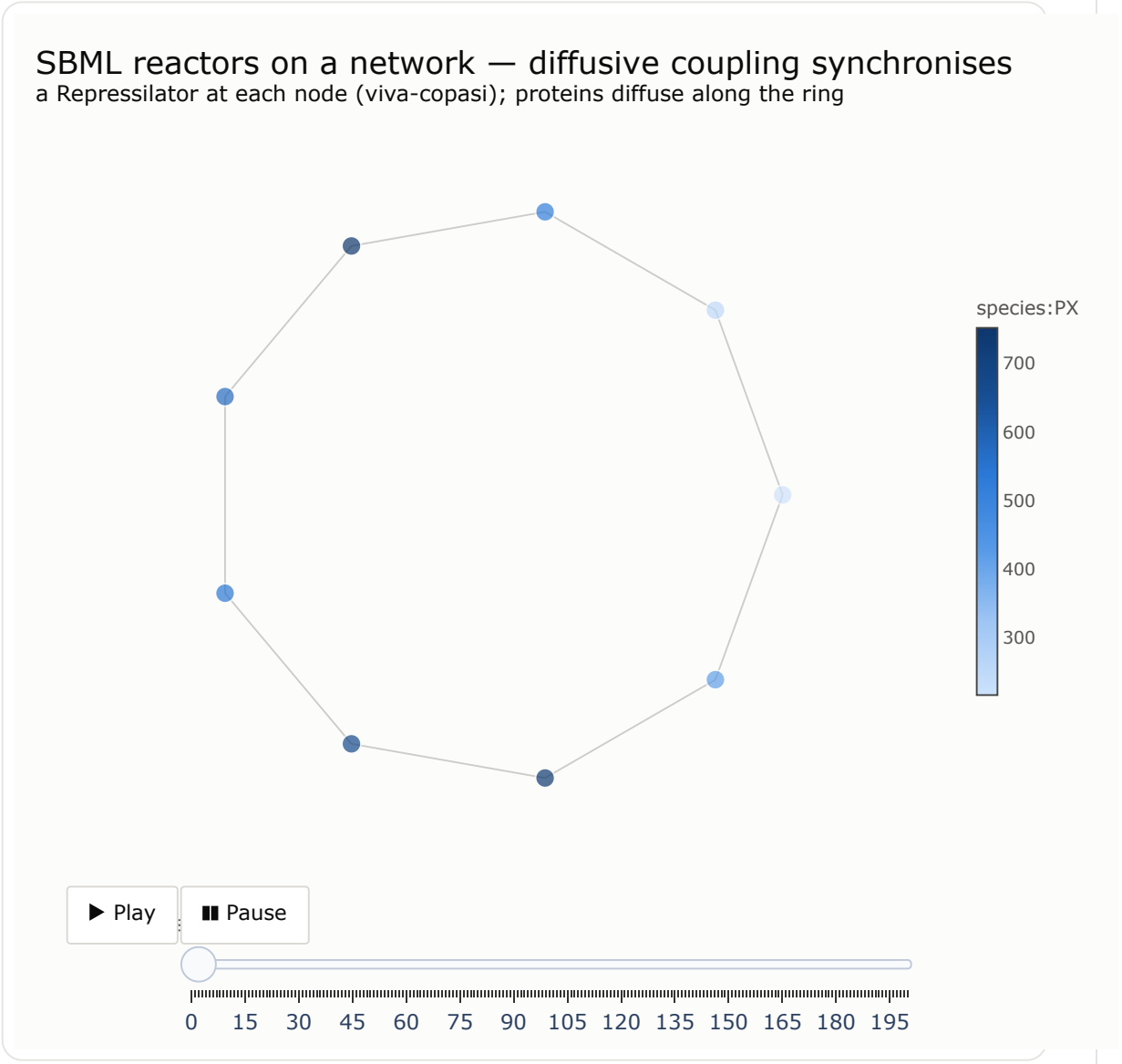
Model & key assumptions

Methods · Simulations

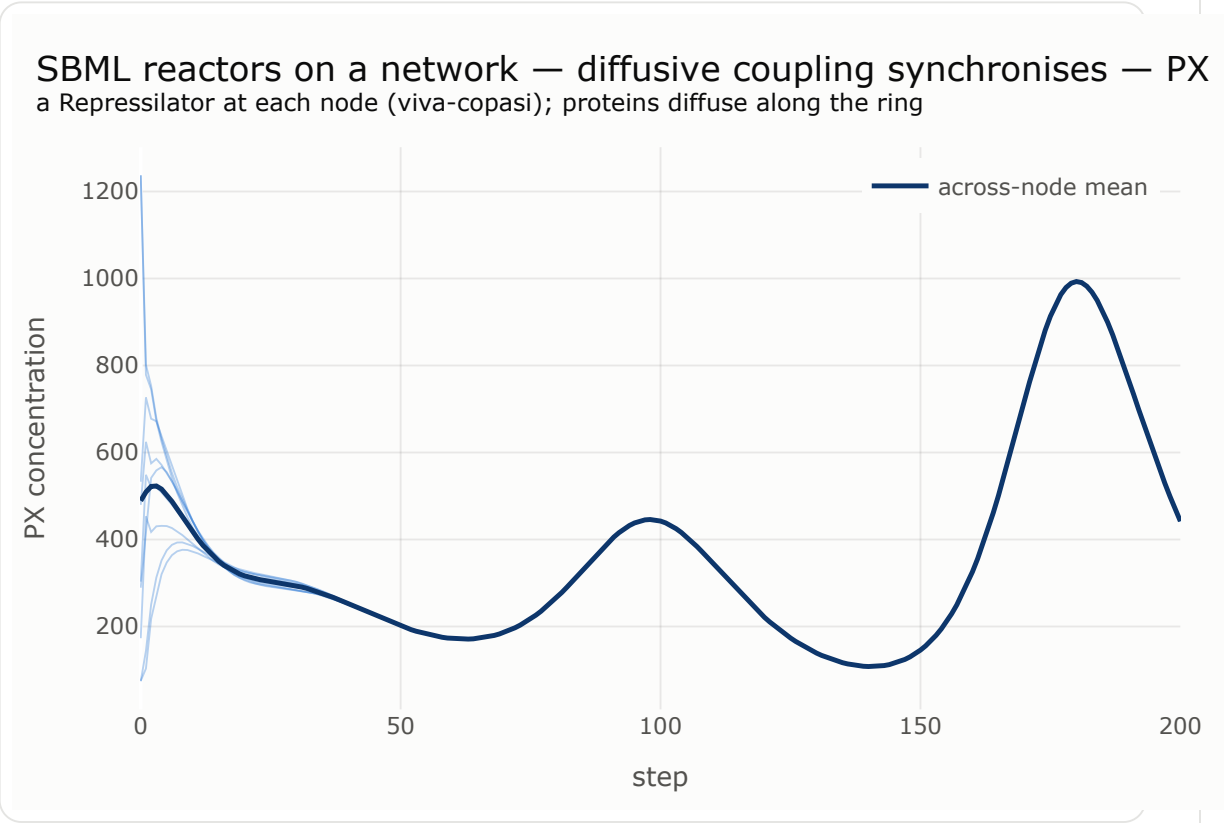
Model `repressilator_network_coupled`

Visualizations

REPRESSILATOR NETWORK COUPLED



REPRESSILATOR NETWORK COUPLED TIMESERIES



Finding

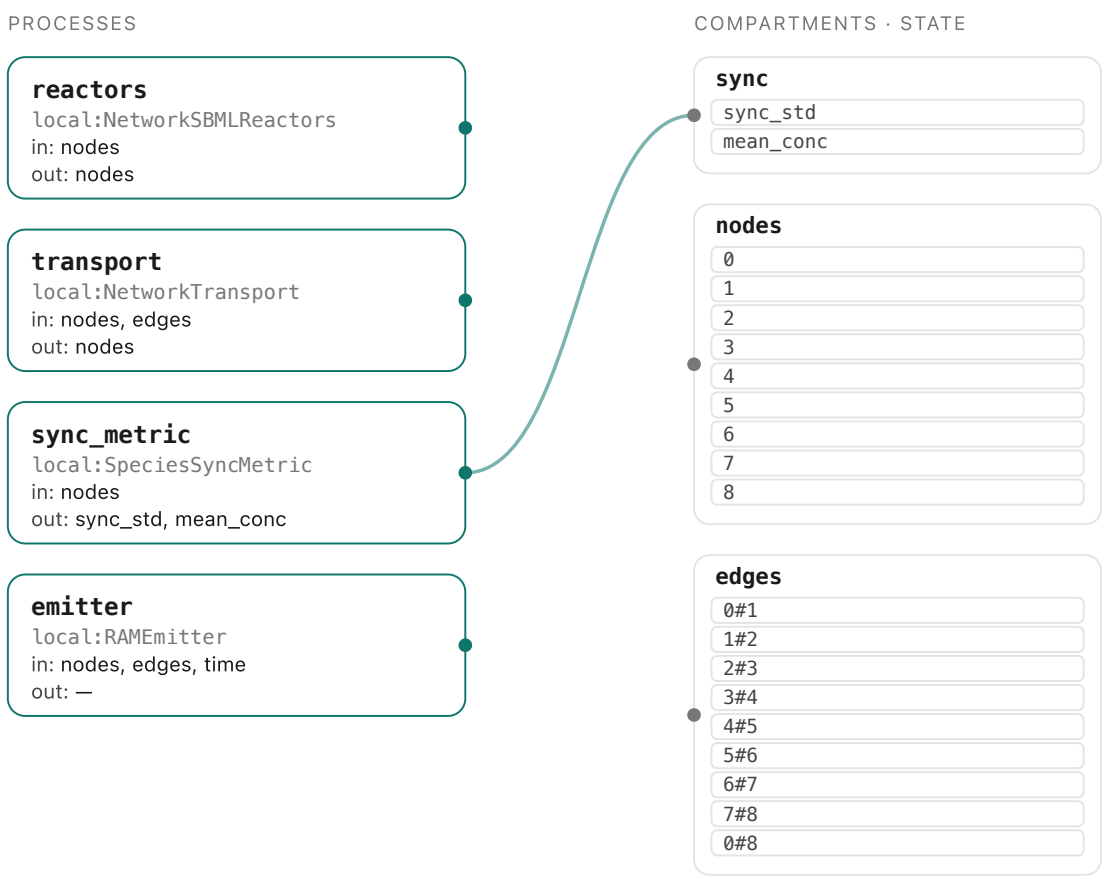
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_coupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_coupled`

Simulation run

Reproduce: `vwb run study 09-sbml-nodes-coupled`

Tests

Final across-node PX spread is <25% of its initial value.
`coupled-synchronises · quantitative · sync_std`

`at_most 0.25` pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 10

SBML reactors on a network — uncoupled control (no diffusion)

● ACCEPTED

With diffusion switched off, do the per-node oscillators stay out of phase?

Hypothesis & discriminating control

Without inter-node transport each Repressilator runs independently, so nodes seeded at different phases never synchronise and the across-node spread stays

large.

MODEL EXPERIMENT Run the repressilator_network_uncoupled composite (SBML reactors + inter-node transport) and measure across-node synchrony.

OBSERVABLE `1 acceptance test`

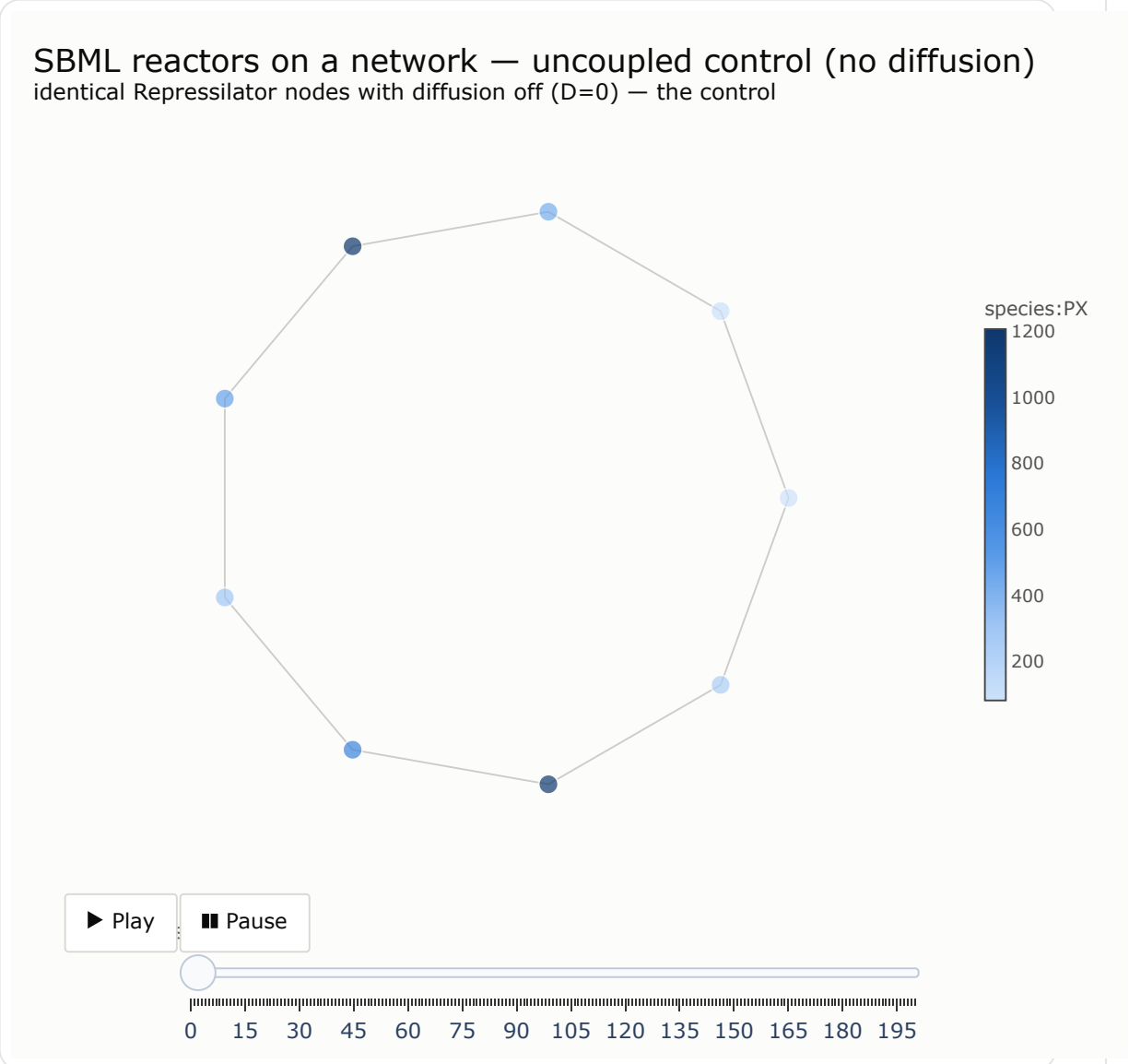
Model & key assumptions

Methods · Simulations

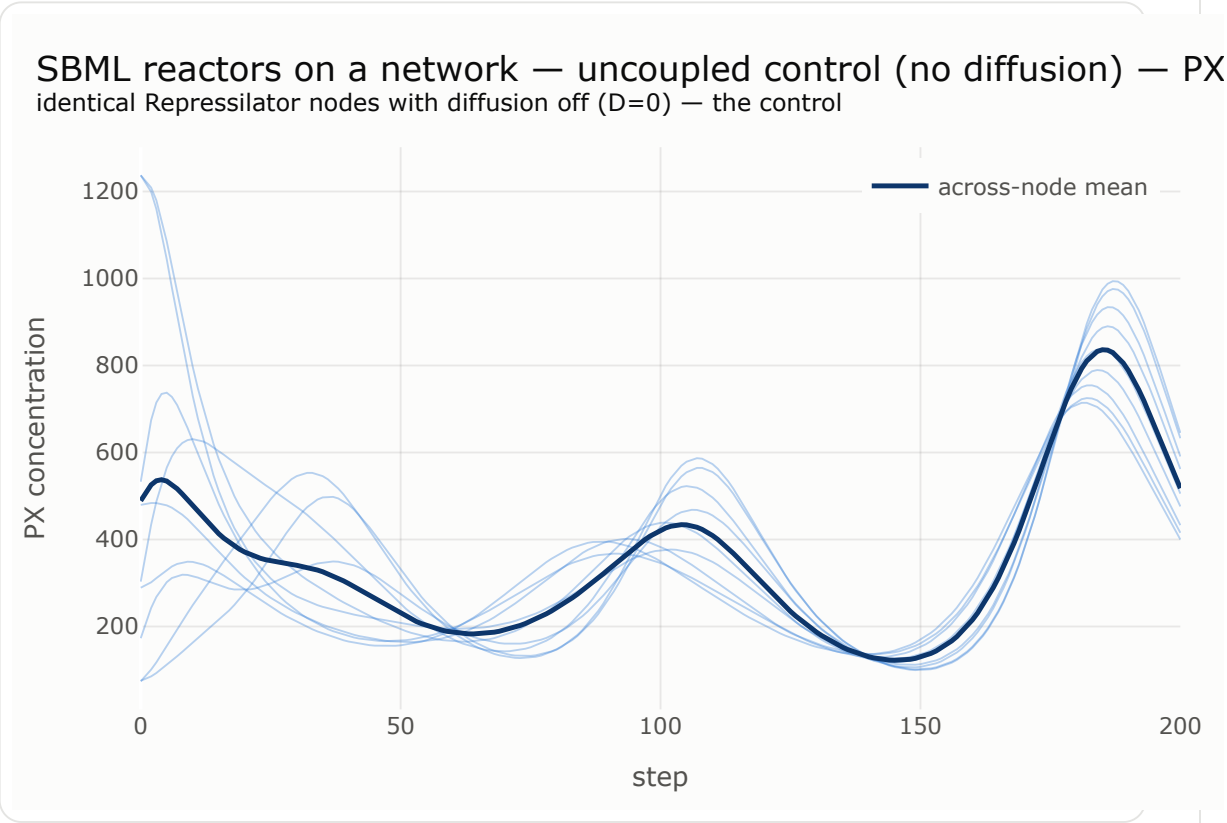
Model `repressilator_network_uncoupled`

Visualizations

REPRESSILATOR NETWORK UNCOUPLED



REPRESSILATOR NETWORK UNCOUPLED TIMESERIES



Finding

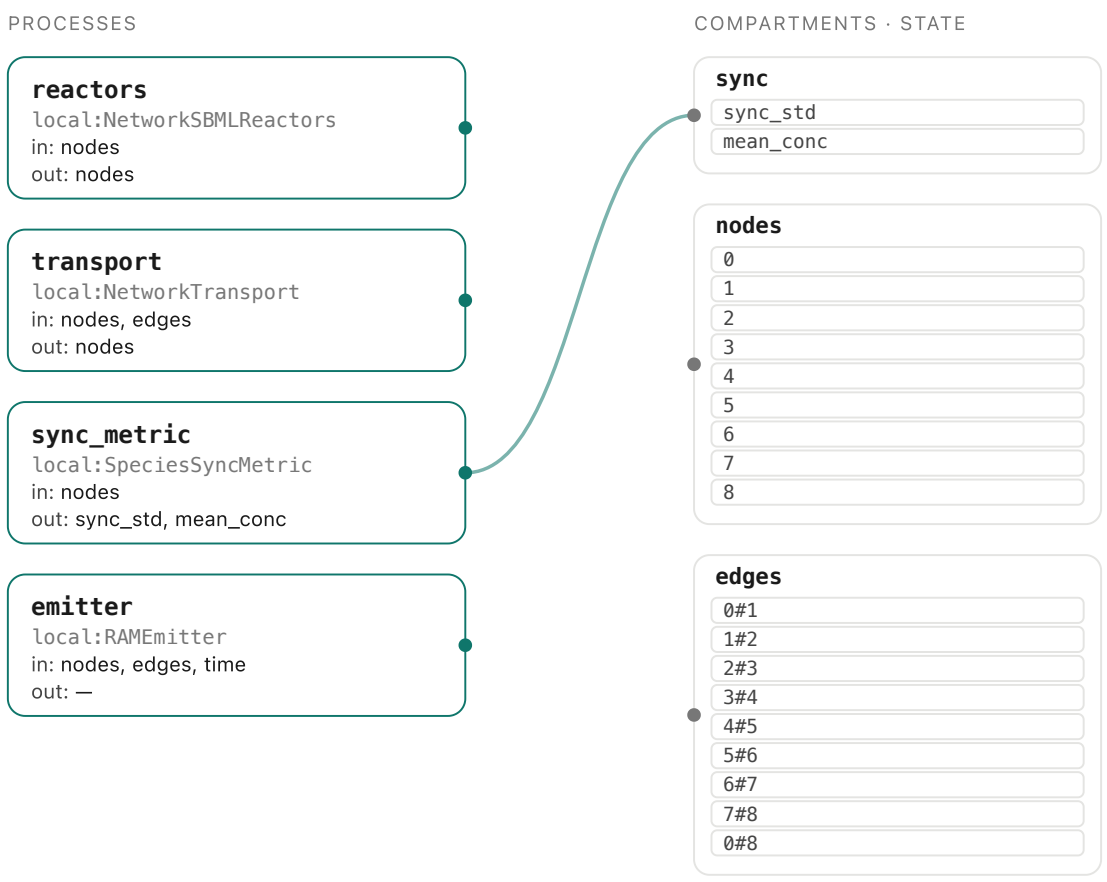
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_uncoupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_uncoupled`

Simulation run

Reproduce: `vwb run study 10-sbml-nodes-uncoupled`

Tests

Final across-node PX spread stays well above the coupled case (ratio not collapsed). `at_least 3` pending

`uncoupled-stays-spread · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

11 Dynamics on a growing network — reactors entrain as the graph is built

● ACCEPTED

Can SBML reactor dynamics run on a network whose topology is itself being built by a process-bigraph reaction, and do newcomers entrain?

Hypothesis & discriminating control

Running the Family-B reactors + transport on a graph that a Family-A growth reaction is simultaneously building, each newcomer joins at its own phase and is pulled into synchrony by the dense local coupling — the full synthesis of dynamics OF and ON the network.

MODEL EXPERIMENT Run the growing_repressilator_network composite — SBML reactors + transport on a topology that a PBG reaction reconfigures during the run — and measure synchrony.

OBSERVABLE `1 acceptance test`

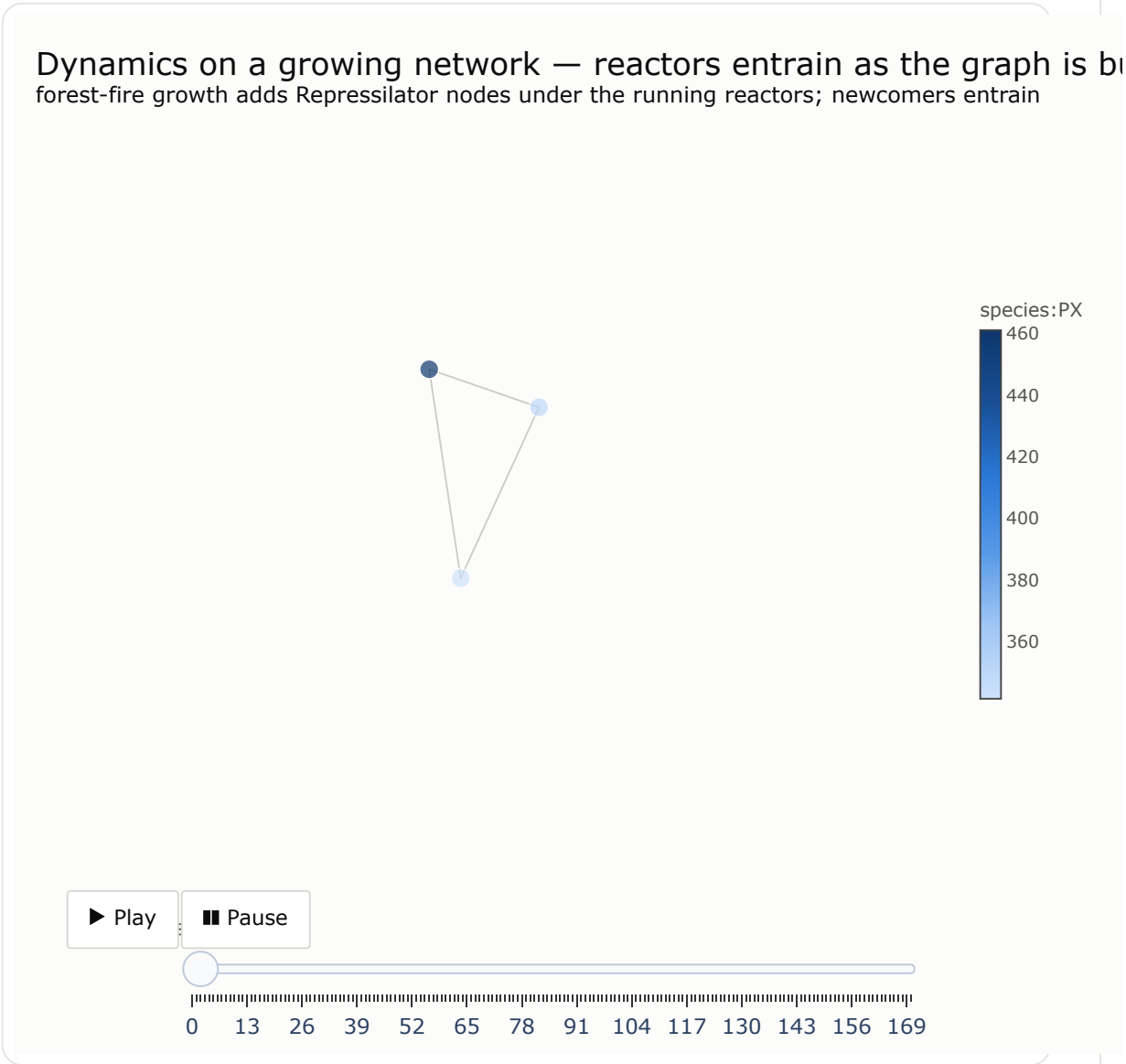
Model & key assumptions

Methods · Simulations

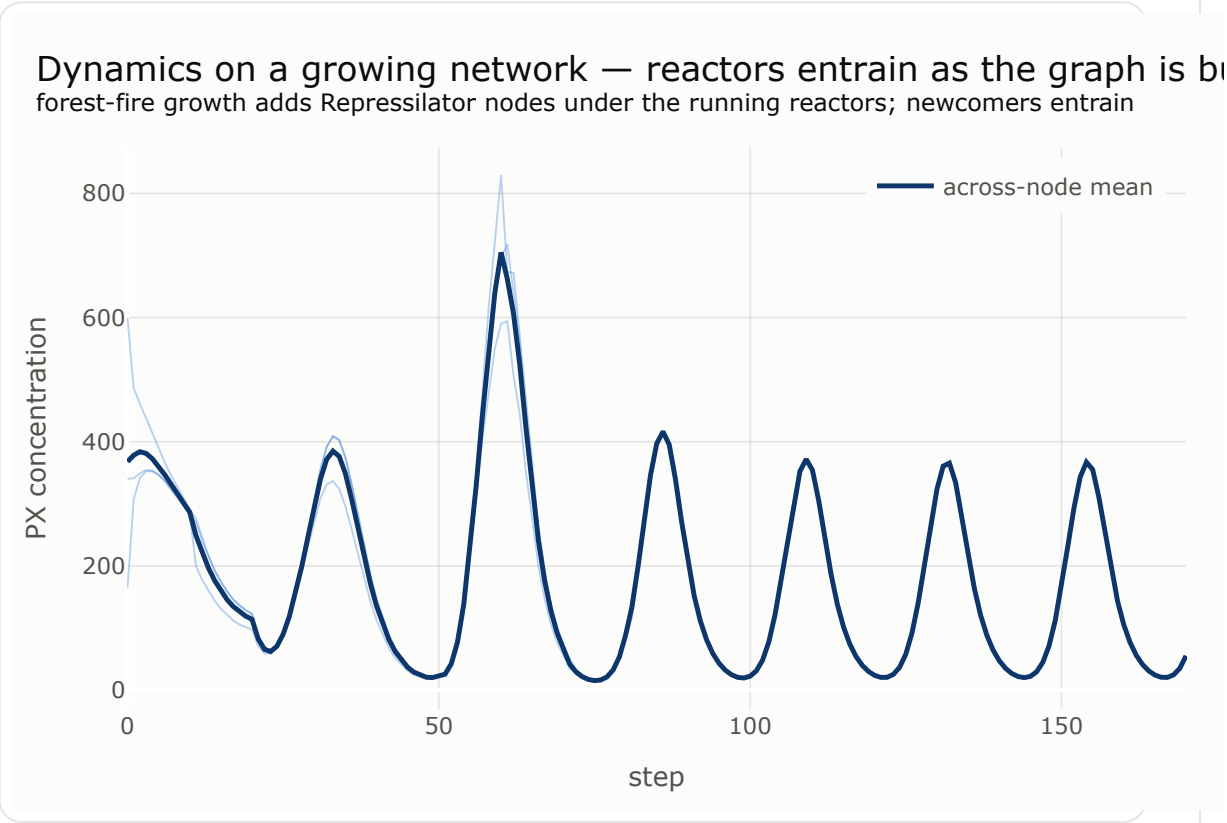
Model `growing_repressilator_network`

Visualizations

GROWING REPRESSILATOR NETWORK



GROWING REPRESSILATOR NETWORK TIMESERIES



Finding

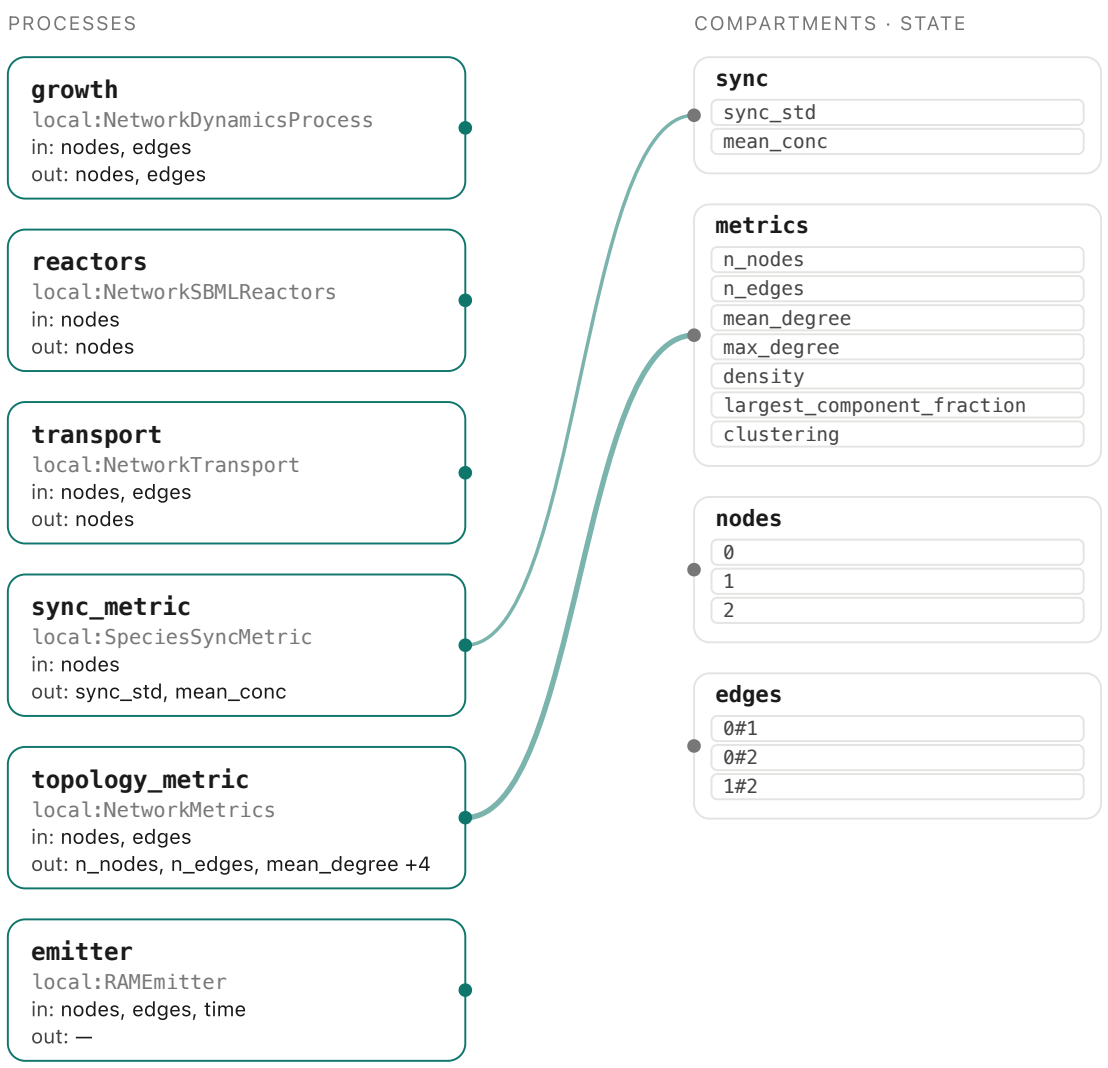
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_c.growing_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.growing_repressilator_network`

Simulation run

Reproduce: `vwb run study 11-coevolving-growing-network`

Tests

Across-node PX spread stays well below the oscillation amplitude even as the network grows (most nodes entrained). `at_most 150` pending

`grows-and-mostly-entrains · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

12 Dynamics on a rewiring network — synchrony survives continual rewiring

ACCEPTED

Do the coupled oscillators stay synchronised when the very edges that couple them are continuously rewired?

Hypothesis & discriminating control

With the topology continuously small-world-rewired under the running reactors (node and edge counts conserved), the coupling remains strong enough to reach and hold synchrony.

MODEL EXPERIMENT Run the rewiring_repressilator_network composite — SBML reactors + transport on a topology that a PBG reaction reconfigures during the run — and measure synchrony.

OBSERVABLE `1 acceptance test`

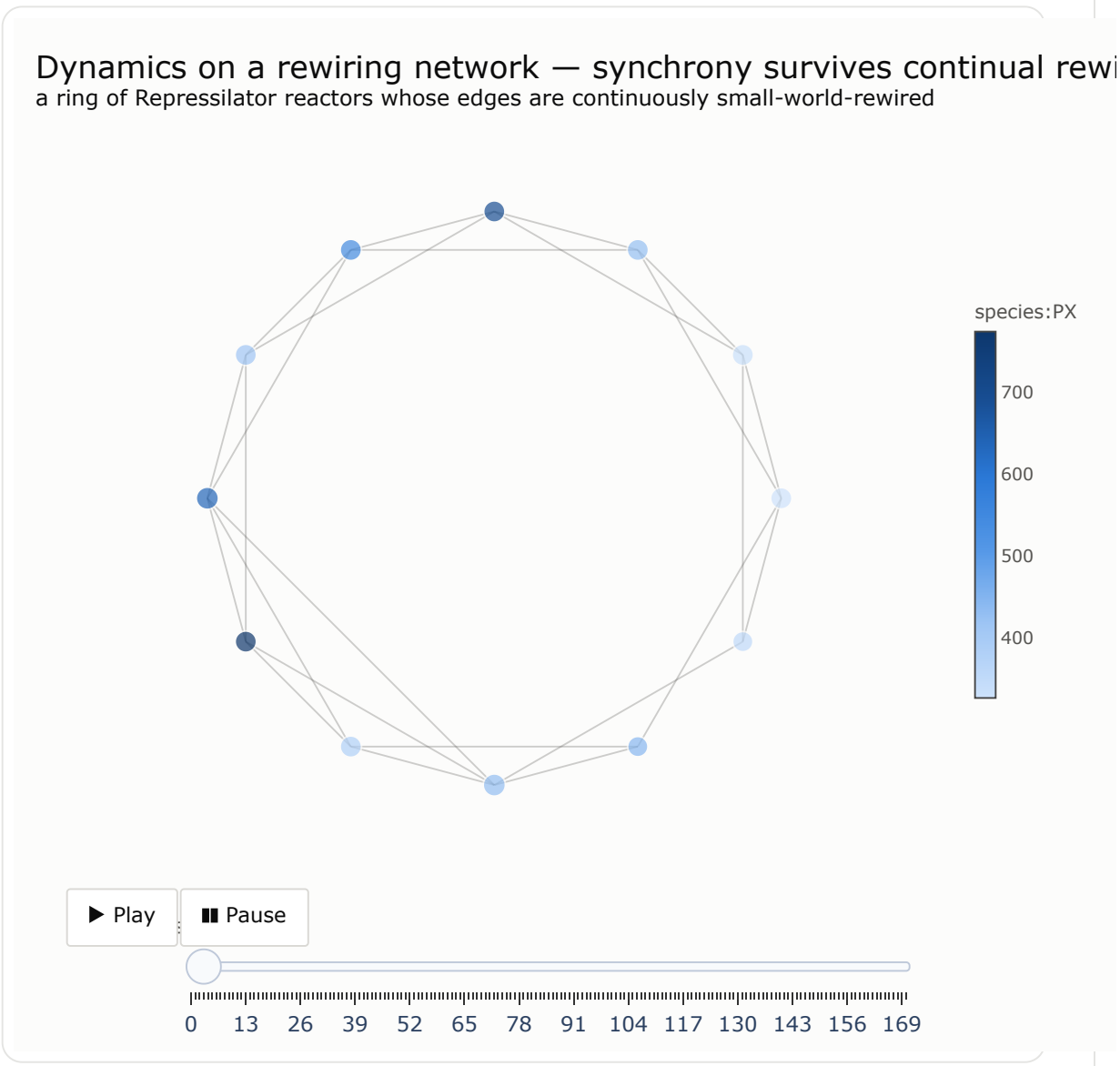
Model & key assumptions

Methods · Simulations

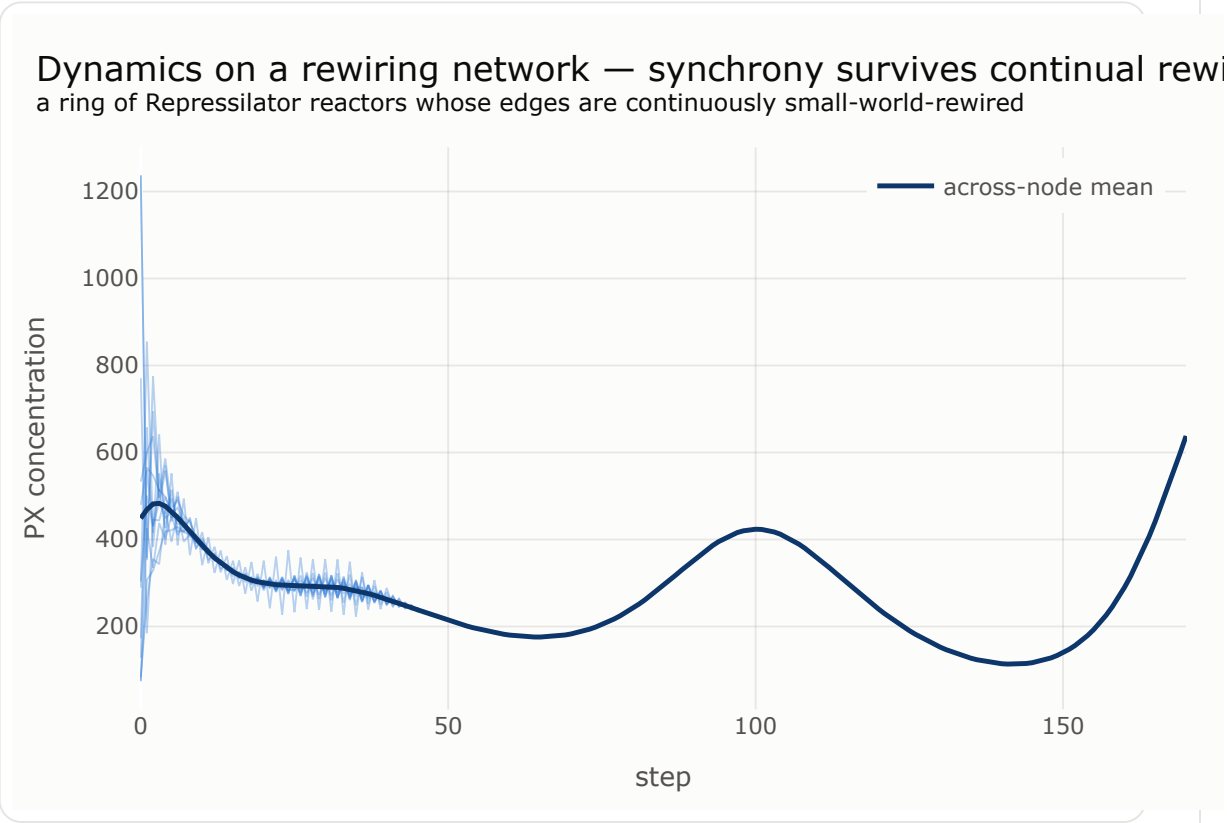
Model `rewiring_repressilator_network`

Visualizations

REWIRING REPRESSILATOR NETWORK



REWIRING REPRESSILATOR NETWORK TIMESERIES



Finding

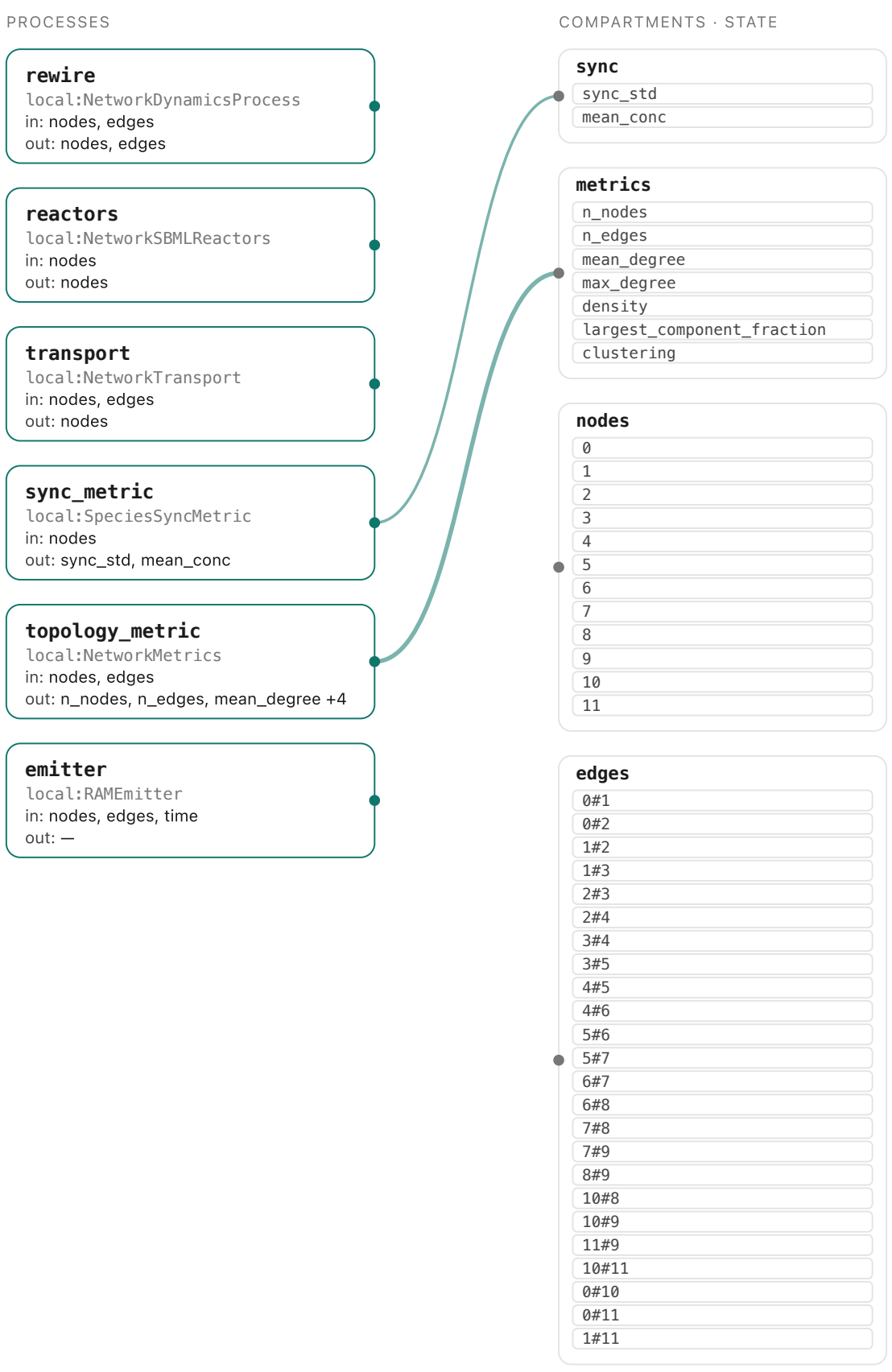
finding · Plausible

Verdict

PASSED **Gate cleared** · Accepted

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_c.rewiring_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.rewiring_repressilator_network`

Simulation run

Reproduce: `vwb run study 12-coevolving-rewiring-network`

Tests

Final across-node PX spread is <30% of the early value despite continual rewiring. `at_most 0.3` pending

`rewires-and-synchronises · quantitative · sync_std`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 13 Quorum-sensing coupling (proposed) ● PLANNED

Does coupling through a single diffusible signal molecule synchronise the networked reactors as effectively as diffusing the internal proteins?

Hypothesis & discriminating control

A dedicated autoinducer species produced and sensed per node, diffusing along the edges, provides indirect coupling sufficient for synchronisation at lower transport flux than direct protein diffusion.

MODEL EXPERIMENT Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09.

Model & key assumptions

Methods · Simulations

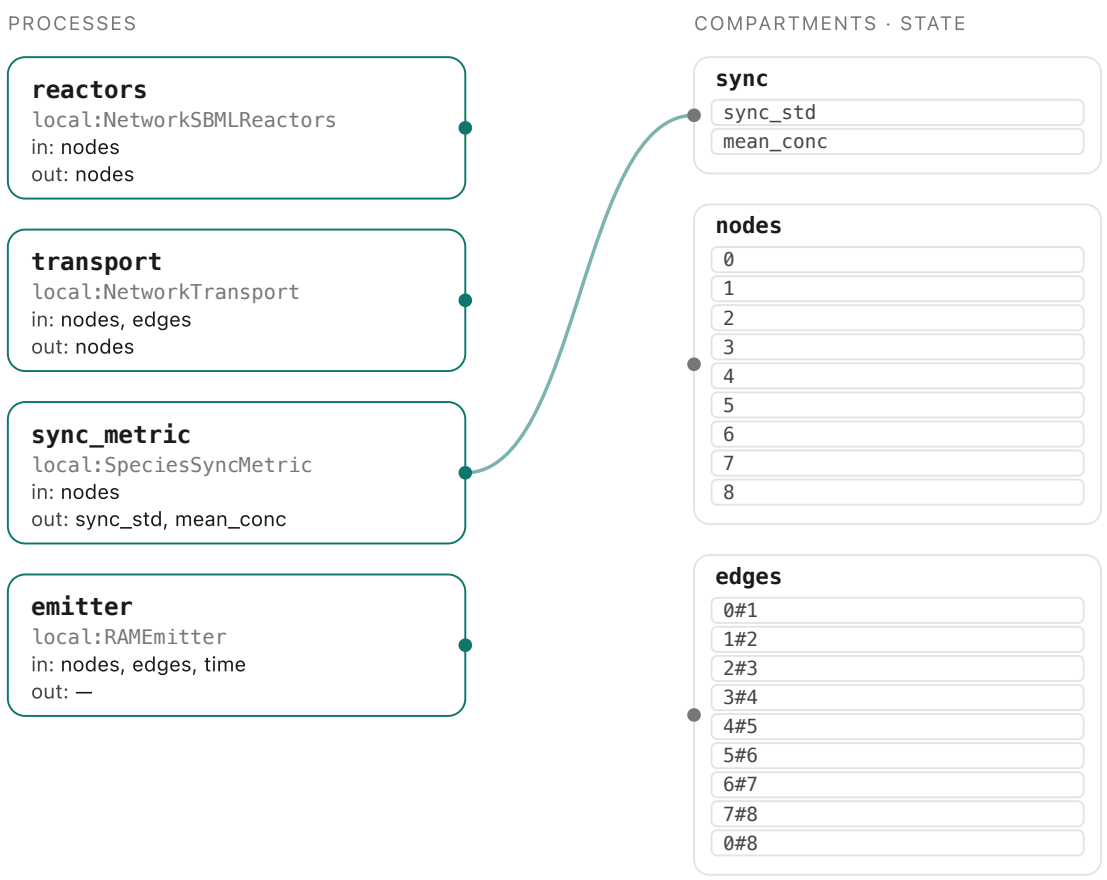
Model `repressilator_network_coupled`

Verdict

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_b.repressilator_network_coupled` — 4 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_b.repressilator_network_coupled`

Simulation run

Reproduce: `vwb run study 13-quorum-sensing-coupling`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 14 Model-agnostic node reactor — Brusselator (proposed)

● PLANNED

Is the SBML-at-nodes architecture model-agnostic — does swapping the Repressilator for a Brusselator reproduce reaction–diffusion patterning on the network with no code change?

Hypothesis & discriminating control

Because NetworkSBMLReactors loads any SBML model and NetworkTransport diffuses any species, pointing them at a Brusselator yields Turing-like patterning / synchronisation on the network with the identical process code.

MODEL EXPERIMENT Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence.

Model & key assumptions

Methods · Simulations

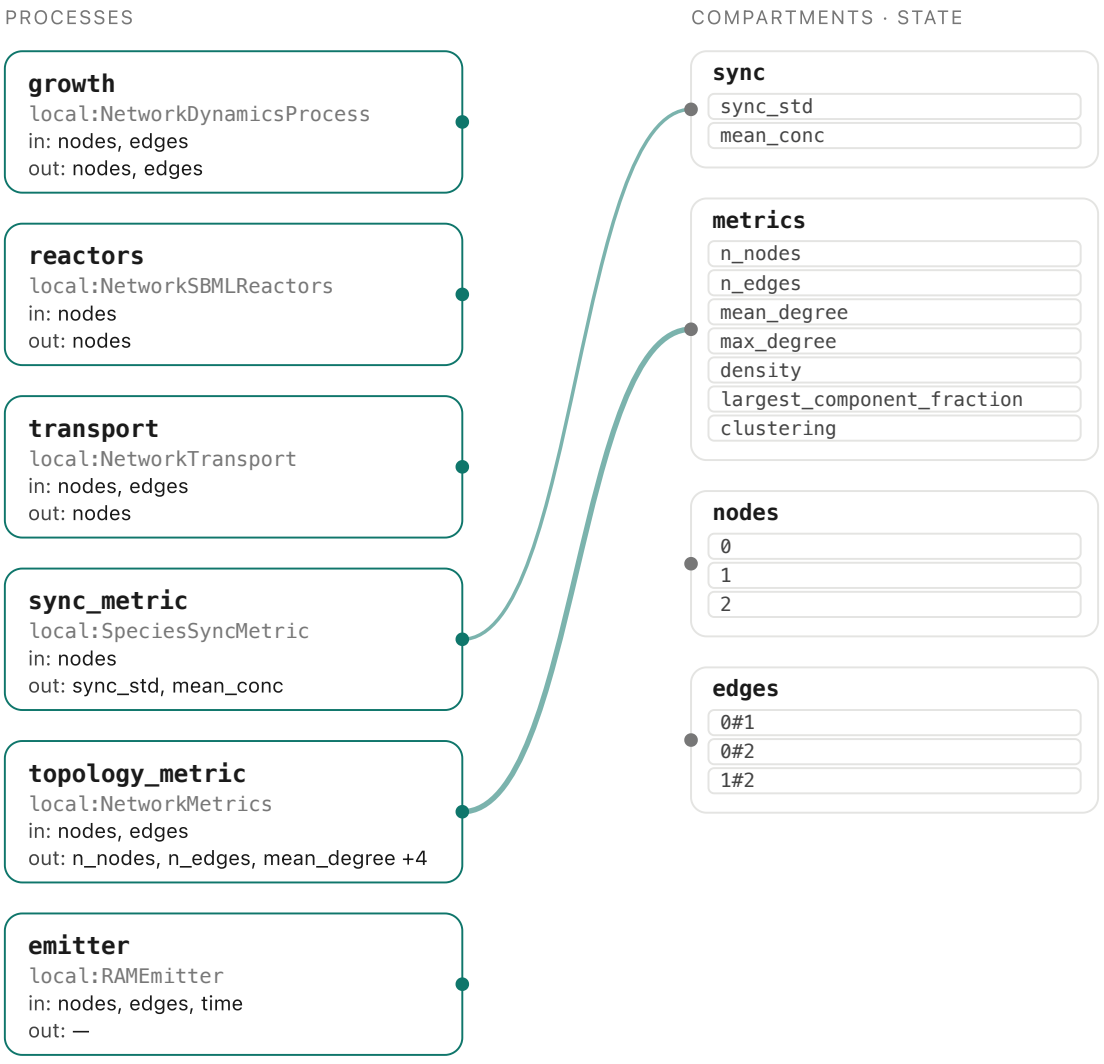
Model `growing_repressilator_network`

Verdict

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_dynamic_networks.composites.family_c.growing_repressilator_network` — 6 processes wired to 4 compartments over 2 links. ● writes · ● reads.

`viva_dynamic_networks.composites.family_c.growing_repressilator_network`

Simulation run

Reproduce: `vwb run study 14-brusselator-model-agnostic`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 15 SBML-driven adaptive coevolution (proposed)

● PLANNED

When the continuous node state (an SBML readout) itself drives the rewiring, does a self-organised modular topology emerge?

Hypothesis & discriminating control

Letting each node rewire toward neighbours with a similar SBML observable (e.g. oscillator phase) closes the state→topology feedback and self-organises the network into phase-coherent modules.

MODEL EXPERIMENT Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07).

Model & key assumptions

Methods · Simulations

Model `adaptive_rewire`

Verdict

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

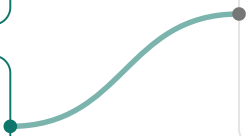
Model — composite internals

PROCESSES

network
local:NetworkDynamicsProcess
in: nodes, edges
out: nodes, edges

metrics_process
local:NetworkMetrics
in: nodes, edges
out: n_nodes, n_edges, mean_degree +4

emitter
local:RAMEmitter
in: nodes, edges, time
out: —



COMPARTMENTS · STATE

metrics

- n_nodes
- n_edges
- mean_degree
- max_degree
- density
- largest_component_fraction
- clustering

nodes

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
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- 37
- 38
- 39

edges

- 0#1
- 0#2
- 1#2
- 1#3
- 2#3
- 2#4
- 3#4
- 3#5
- 4#5
- 4#6
- 5#6
- 5#7
- 6#7
- 6#8

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Static bigraph of `viva_dynamic_networks.composites.family_a.adaptive_rewire` — 3 processes wired to 3 compartments over 1 link. ● writes · ● reads.

`viva_dynamic_networks.composites.family_a.adaptive_rewire`

Simulation run

Reproduce: `vwb run study 15-sbml-adaptive-coevolution`

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

Contradictions & uncertainty

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

Erdős–Rényi random growth

Next studies & priorities

P1 Quorum-sensing coupling (proposed)

Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09.

— 13-quorum-sensing-coupling

P1 Model-agnostic node reactor — Brusselator (proposed)

Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence.

— 14-brusselator-model-agnostic

P1 SBML-driven adaptive coevolution (proposed)

Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07).

— 15-sbml-adaptive-coevolution

OPEN EXPERT DECISIONS (1)

1. Build out the proposed follow-ups (studies 13–15)?

Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

✓ No open validation flags — the deterministic scan found nothing to act on.

▼ REPORT READINESS — 152 FINDINGS ACROSS 14 CHECKS (27 ERRORS, 125 WARNINGS)

12 finding without statement error

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 visualization class misplaced error

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

3 missing visualizations error

`13-quorum-sensing-coupling, 14-brusselator-model-agnostic, 15-sbml-adaptive-coevolution`

20 gate class missing warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

15 status legacy only warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +11`

14 dag edges legacy only warning

`02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation, 05-duplication-divergence +10`

12 missing conditions block warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 missing planned runs warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 missing readouts warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 status claims done no runs recorded warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 missing simulation set warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

12 missing variants warning

`01-erdos-renyi-random-growth, 02-preferential-attachment, 03-small-world-rewiring, 04-fitness-condensation +8`

3 visualization gap warning

`13-quorum-sensing-coupling, 14-brusselator-model-agnostic, 15-sbml-adaptive-coevolution`

1 investigation missing adversarial warning

`dynamic-networks`

Claim–evidence map

Each study's central scientific claim, its evidence status, and a link to the study.

CLAIM	Status	Study
Random growth produces a homogeneous, hub-free network (Poisson-like degrees) — the null baseline the structured rules are read against.	SUPPORTED	Erdős–Rényi random growth
Degree-proportional attachment produces scale-free hubs with a power-law degree tail — unlike random growth.	SUPPORTED	Barabási–Albert preferential attachment
Random rewiring of a ring lattice preserves edge count while shortcuts collapse path length — producing the small-world regime.	SUPPORTED	Watts–Strogatz small-world rewiring
Fitness-weighted attachment still builds hubs, but a high-fitness latecomer can overtake older nodes — unlike pure Barabási–Albert.	SUPPORTED	Bianconi–Barabási fitness attachment
Copy-then-diverge growth reproduces the heavy-tailed, locally clustered topology seen in protein-interaction networks.	SUPPORTED	Duplication–divergence (biological growth)
Forest-fire growth densifies the network: mean degree rises as the graph grows, reproducing the densification law.	SUPPORTED	Forest-fire growth & densification
Coevolving opinions and rewiring self-organise the network into like-minded clusters — topology and state adapt to each other.	SUPPORTED	Adaptive coevolution (opinions + rewiring)
Activity-driven firing produces a sparse, blinking temporal network — each snapshot is instantaneous, unlike the accreting Family-A graphs.	SUPPORTED	Activity-driven temporal network
With an SBML Repressilator at each node and protein diffusion along the edges, the coupled oscillators synchronise — across-node concentration spread collapses.	SUPPORTED	SBML reactors on a network — diffusive coupling synchronises
With diffusion off, the per-node oscillators remain independent and out of phase — the control that isolates coupling as the cause of synchronisation in study 09.	SUPPORTED	SBML reactors on a network — uncoupled control (no diffusion)
Continuous SBML dynamics run correctly on a network being built node-by-node by a PBG reaction: the system stays bounded and oscillating, and most newcomers entrain through the dense local coupling — but synchronisability is topology-dependent, with weakly-attached (low-degree) nodes lagging the syn	SUPPORTED	Dynamics on a growing network — reactors entrain as the graph is built
Coupled SBML oscillators synchronise and stay synchronised even as the coupling edges are continuously rewired — reaction–diffusion is robust to an evolving (but well-connected) topology.	SUPPORTED	Dynamics on a rewiring network — synchrony survives continual rewiring

Evidence register

Every study in the campaign (15) — scientific verdict, run status, and one-line result, each linked to its full study card above.

STUDY	Verdict	Run	Result
Erdős–Rényi random growth	SUPPORTED	ran	Random growth produces a homogeneous, hub-free network (Poisson-like degrees) — the null baseline the structured rules are read against.

STUDY	Verdict	Run	Result
Barabási–Albert preferential attachment	SUPPORTED	ran	Degree-proportional attachment produces scale-free hubs with a power-law degree tail — unlike random growth.
Watts–Strogatz small-world rewiring	SUPPORTED	ran	Random rewiring of a ring lattice preserves edge count while shortcuts collapse path length — producing the small-world regime.
Bianconi–Barabási fitness attachment	SUPPORTED	ran	Fitness-weighted attachment still builds hubs, but a high-fitness latecomer can overtake older nodes — unlike pure Barabási–Albert.
Duplication–divergence (biological growth)	SUPPORTED	ran	Copy-then-diverge growth reproduces the heavy-tailed, locally clustered topology seen in protein-interaction networks.
Forest-fire growth & densification	SUPPORTED	ran	Forest-fire growth densifies the network: mean degree rises as the graph grows, reproducing the densification law.
Adaptive coevolution (opinions + rewiring)	SUPPORTED	ran	Coevolving opinions and rewiring self-organise the network into like-minded clusters — topology and state adapt to each other.
Activity-driven temporal network	SUPPORTED	ran	Activity-driven firing produces a sparse, blinking temporal network — each snapshot is instantaneous, unlike the accreting Family-A graphs.
SBML reactors on a network — diffusive coupling synchronises	SUPPORTED	ran	With an SBML Repressilator at each node and protein diffusion along the edges, the coupled oscillators synchronise — across-node concentration spread collapses.
SBML reactors on a network — uncoupled control (no diffusion)	SUPPORTED	ran	With diffusion off, the per-node oscillators remain independent and out of phase — the control that isolates coupling as the cause of synchronisation in study 09.
Dynamics on a growing network — reactors entrain as the graph is built	SUPPORTED	ran	Continuous SBML dynamics run correctly on a network being built node-by-node by a PBG reaction: the system stays bounded and oscillating, and most newcomers entrain through the dense local coupling — but synchronisabilit
Dynamics on a rewiring network — synchrony survives continual rewiring	SUPPORTED	ran	Coupled SBML oscillators synchronise and stay synchronised even as the coupling edges are continuously rewired — reaction–diffusion is robust to an evolving (but well-connected) topology.
Quorum-sensing coupling (proposed)	PENDING	planned	Extend the node SBML model with a signaling species and diffuse only that molecule; compare synchronisation to study 09.
Model-agnostic node reactor — Brusselator (proposed)	PENDING	planned	Vendor a Brusselator SBML, point the reactors at it, and reproduce a Family-B/C study to demonstrate model-independence.
SBML-driven adaptive coevolution (proposed)	PENDING	planned	Couple an SBML-observable-driven rewiring rule to the Family-B reactors, so topology and continuous state co-adapt (the continuous generalisation of study 07).

References

The papers behind each acceptance band, finding and literature anchor.

Elowitz MB 2000 `elowitz2000repressilator` — A synthetic oscillatory network of transcriptional regulators · Elowitz MB, Leibler S, 2000 · source ↗ 4×

Leskovec J 2007 `leskovec2007graph` — Graph evolution: Densification and shrinking diameters · Leskovec J, Kleinberg J, Faloutsos C, 2007 · source ↗ 3×

Watts DJ 1998 `watts1998smallworld` — Collective dynamics of 'small-world' networks · Watts DJ, Strogatz SH, 1998 · source ↗ 3×

Albert R 2002 `albert2002statistical` — Statistical mechanics of complex networks · Albert R, Barabási A-L, 2002 · source ↗ 2×

Barabási A-L 1999 `barabasi1999emergence` — Emergence of scaling in random networks · Barabási A-L, Albert R, 1999 · source ↗ 2×

Bianconi G 2001 `bianconi2001competition` — Competition and multiscaling in evolving networks · Bianconi G, Barabási A-L, 2001 · source ↗ 2×

Colizza V 2007 `colizza2007reaction` — Reaction–diffusion processes and metapopulation models in heterogeneous networks · Colizza V, Pastor-Satorras R, Vespignani A, 2007 · source ↗ 2×

Erdős P 1960 `erdos1960evolution` — On the evolution of random graphs · Erdős P, Rényi A, 1960 2×

Gross T 2008 `gross2008adaptive` — Adaptive coevolutionary networks: a review · Gross T, Blasius B, 2008 · source ↗ 2×

Ispolatov I 2005 `ispolatov2005duplication` — Duplication-divergence model of protein interaction network · Ispolatov I, Krapivsky PL, Yuryev A, 2005 · source ↗ 2×

Perra N 2012 `perra2012activity` — Activity driven modeling of time varying networks · Perra N, Gonçalves B, Pastor-Satorras R, Vespignani A, 2012 · source ↗ 2×

Holme P 2012 `holme2012temporal` — Temporal networks · Holme P, Saramäki J, 2012 · source ↗ 1×

MEJ 2003 `newman2003structure` — The structure and function of complex networks · Newman MEJ, 2003 · source ↗ 1×

Rodrigues FA 2016 `rodrigues2016kuramoto` — The Kuramoto model in complex networks · Rodrigues FA, Peron TKDM, Ji P, Kurths J, 2016 · source ↗ 1×

Vázquez A 2003 `vazquez2003modeling` — Modeling of protein interaction networks · Vázquez A, Flammini A, Maritan A, Vespignani A, 2003 1×

Glossary

reaction (process-bigraph)
A structural rewrite of the bigraph emitted as _add/_remove sentinels on a map port; here it adds/removes network nodes and edges.

degree
Number of edges incident to a node.

scale-free
A network whose degree distribution has a power-law tail — a few very-high-degree hubs.

small-world

High local clustering with short global path lengths.

densification
Edge count growing super-linearly in node count as a network evolves.

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision Add a note…

Rendered from workspace `viva-dynamic-networks` · deterministic from investigation + study YAML.

`investigations/dynamic-networks/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON`